

Ohio River Trail Council

June 4, 2015

Welcome



Dr. Vincent Troia, CEO



Ohio River Trail Council



Dr. Vincent Troia

President & CEO

Ohio River Trail Council

1100 Pennsylvania Ave

Monaca, PA 15061

Phone: 724.728.2625

drvtroia@ohiorivertrail.org



Presentation Overview



Cycling Safety

❖ *PA Vehicle Code
Title 75*

❖ *Bike PA Route A
Fatalities*

— *Ed Quigley*

Presentation Overview



Bicycle Path

❖ *Bike Path Classifications — Vincent Troia*

Presentation Overview



Ohio River Trail

❖ *Phase 1 & 2 Study Objectives*

— John Orndorff

Presentation Overview



Ohio River Trail

❖ *Phase 1 & 2 Study Update*

— Bob Genter

Cycling Safety

Ed Quigley, Director





Cycling Safety

PA Vehicle Code

Title 75 Chapter 35

Cycling Safety – PA Bike Route A



Ed Quigley

- ▶ Ed Quigley, Director
- ▶ Ohio River Trail Council
- ▶ Bike Path Safety, Chairman
- ▶ Ride of Silence Coordinator
- ▶ Email: ed.quigley@gmail.com



Cycling Safety – Title 75 Chapter 35



10

Special Vehicles and Pedestrians

- ▶ Subchapter A. Operation of Pedalcycles
- ▶ Section § 3501. Applicability of traffic laws to pedalcycles.
- ▶ (a) General rule. **Every person riding a pedalcycle upon a roadway shall be granted all of the rights and shall be subject to all of the duties applicable to the driver of a vehicle** by this title, ...

Cycling Safety – Title 75 Chapter 35



11

Special Vehicles and Pedestrians

- ▶ Bicyclists are considered vehicles.
- ▶ Cyclists have the same rights and same rules.



Cycling Safety – House Bill 170



Safe Passing Bill – Bicycle Safety

- ▶ Governor Corbett signed “Safe Passing Bill” HB 170 Into Law on **February 3, 2012**.
- ▶ This bill amends Title 75 (Vehicles) in order to **improve safety for bicyclists** on PA roads, establishing new rules for motorists to follow when they encounter a bicyclist.

Cycling Safety – House Bill 170



Safe Passing Bill – Full Use of lane

- ▶ Bicyclists are permitted **Full Use of a Lane** if that is the **safest** place to be.



Cycling Safety – House Bill 170



Safe Passing Bill



- ▶ If there is only one lane in each direction, bicyclists may use any portion of the lane to avoid hazards on the roadway.

Cycling Safety – House Bill 170



Safe Passing Bill – Four-Foot

- ▶ Every car that passes a bike must give a minimum of 4-feet of clearance between their vehicle and the bicycle at a “prudent reduced speed.”



**4 FEET
PLEASE**



IT'S THE LAW

Cycling Safety – House Bill 170



Safe Passing Bill – Change Lanes

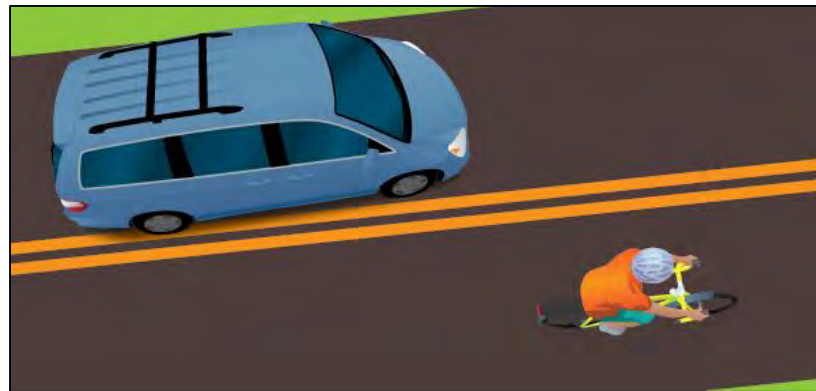


Cycling Safety – House Bill 170



Safe Passing Bill – Double Yellow Line

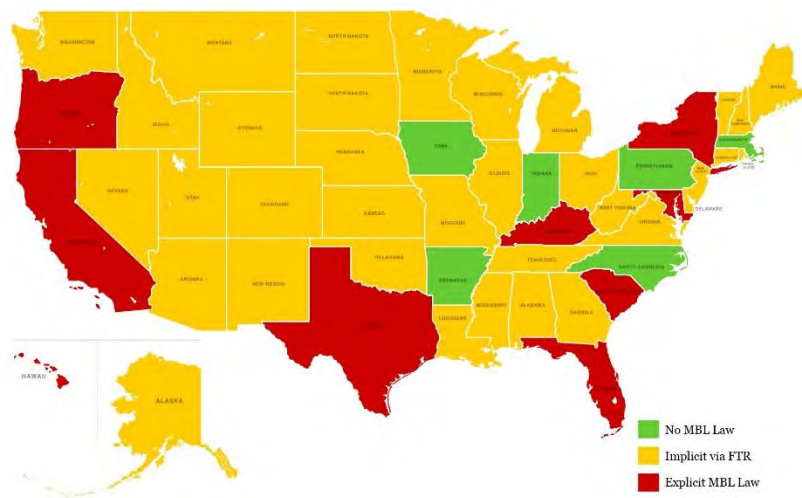
- ▶ Drivers can cross the center double yellow line if necessary as long as it is safe to do so.



Cycling Safety – House Bill 170

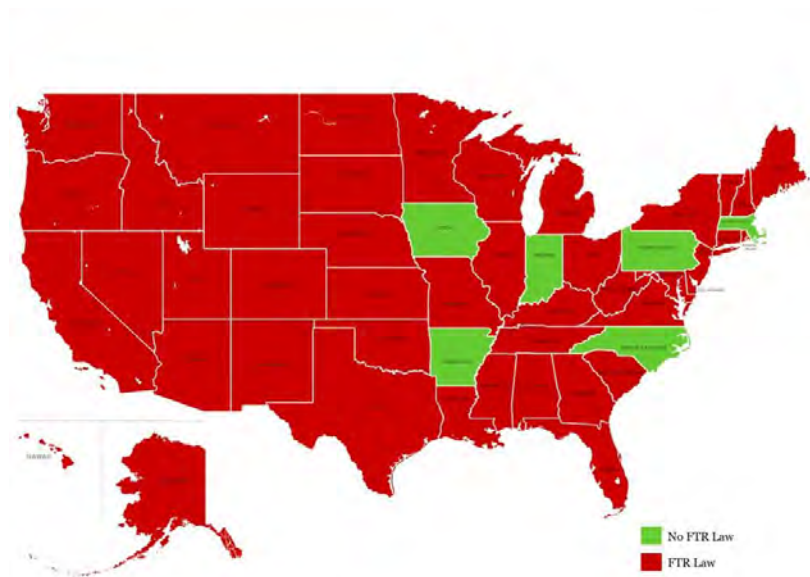


Safe Passing Bill – Bike Lane



- ▶ PA and the six green states do not require the bicyclist to remain in the bike lane.
- ▶ **No Bike Lane Law.**

Safe Passing Bill – Far to the Right

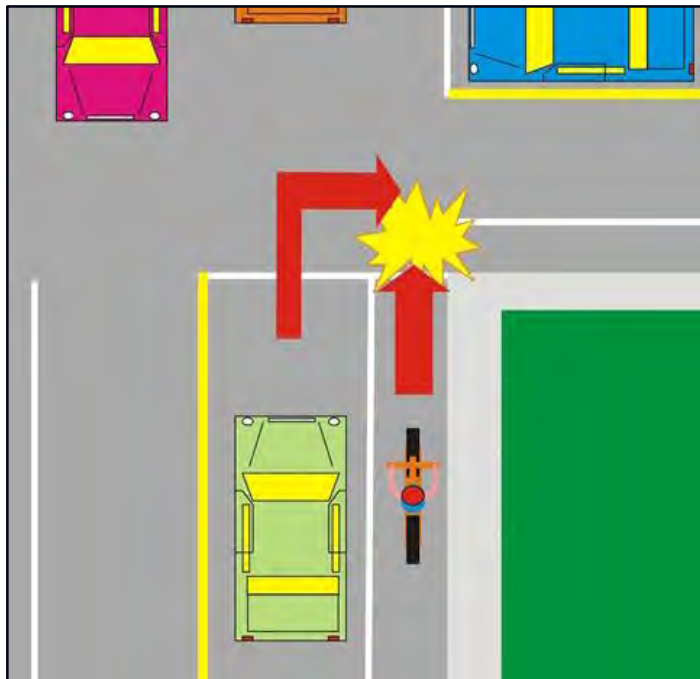


- ▶ PA and the green states have no **Far to the Right** law.
- ▶ PA does not force bicyclists to abandon their **Use Full Lane** rights.

Cycling Safety – House Bill 170



21



Safe Passing Bill

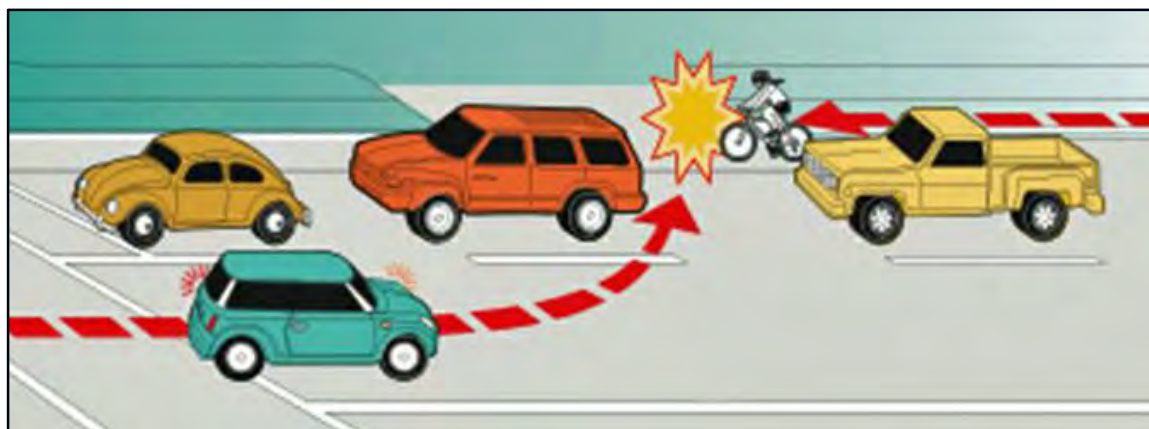
- ▶ No turn by a driver can interfere with a bicyclist who is proceeding straight on a roadway
- ▶ The classic **Right Hook**

Cycling Safety – House Bill 170



Safe Passing Bill - Left Turn

- ▶ Drivers attempting to turn left must yield the right of way to bicyclists traveling in the opposite direction.



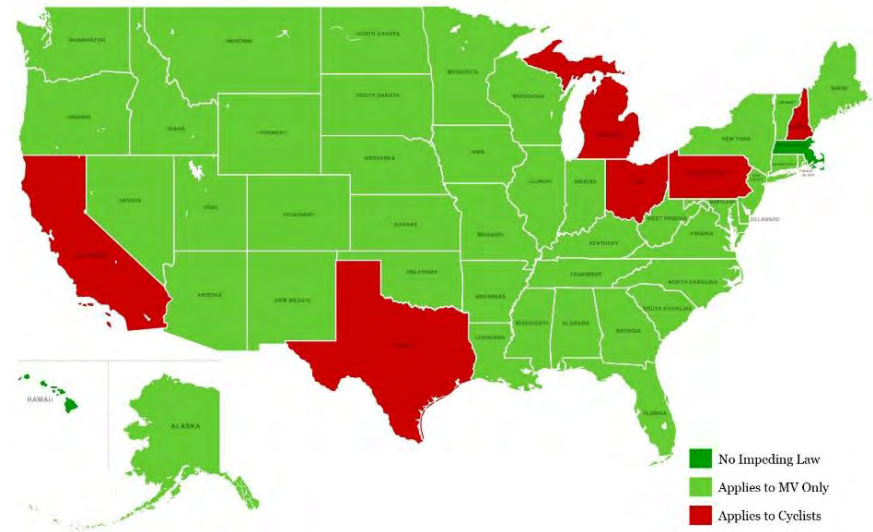
Cycling Safety – House Bill 170



23

Safe Passing Bill - Impeding Law

- ▶ PA and the red states have an Impeding Law for **cyclists** and **motorists**.



Cycling Safety – House Bill 170



Safe Passing Bill - Impeding Law

- ▶ Cyclists and motorists traveling below the posted speed limit must keep as **far to the right** as practical and safe.



Cycling Safety – House Bill 170



Safe Passing Bill - Impeding Law

- ▶ Cyclists exemptions
 - ❖ One-way street.
 - ❖ Making a left turn.



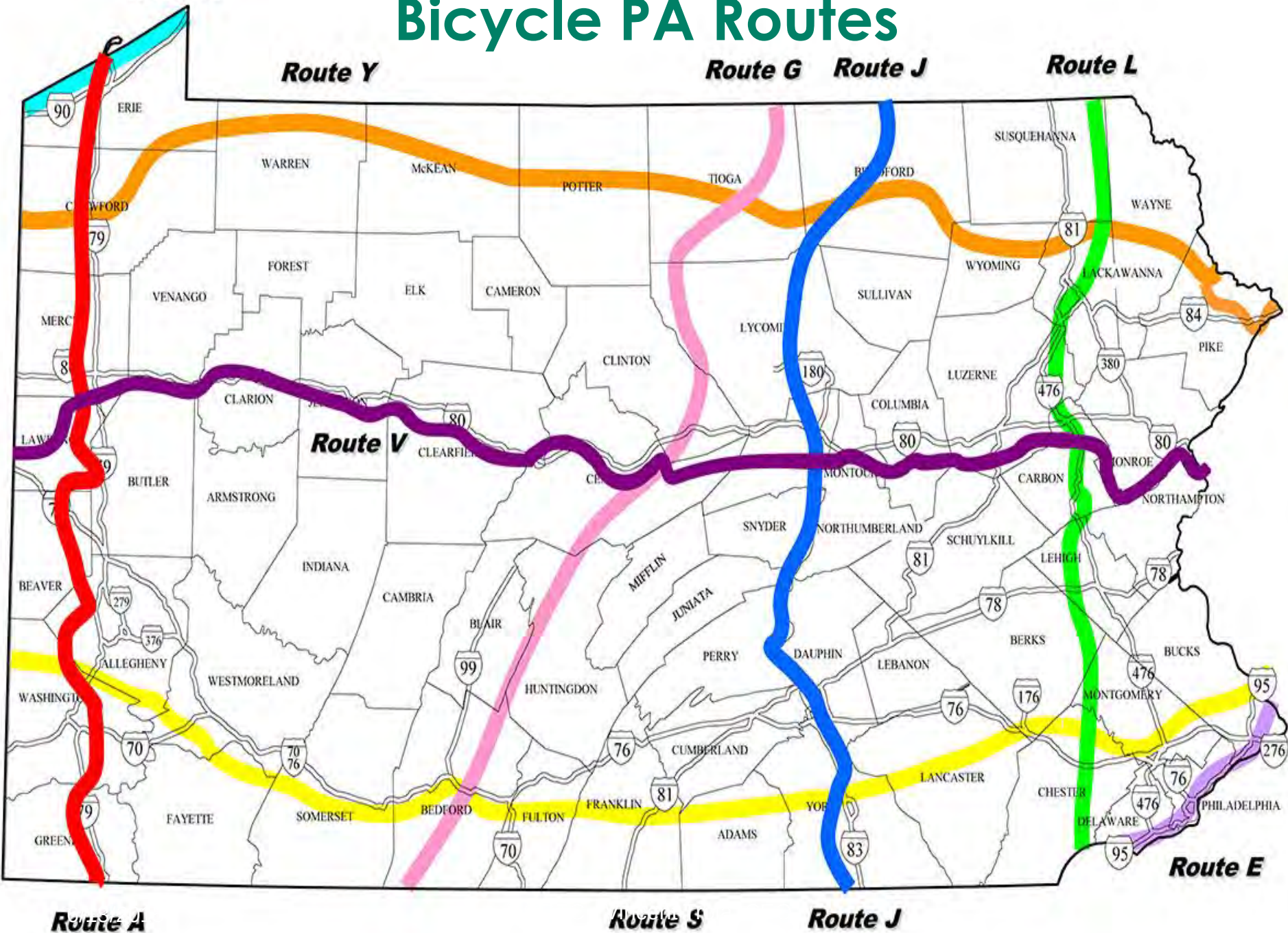


Cycling Safety

PA 51 Bike Route A

Fatalities

Route Z



Bicycle PA Route A

PAGE 10

28



PAGE 9



Bicycle PA Route A

PAGE 8



Scale
1:65,000
0 0.5 1 Miles
0 0.5 1 2 Kilometers

BicyclePA



A

Route



© 2015

1985

2015



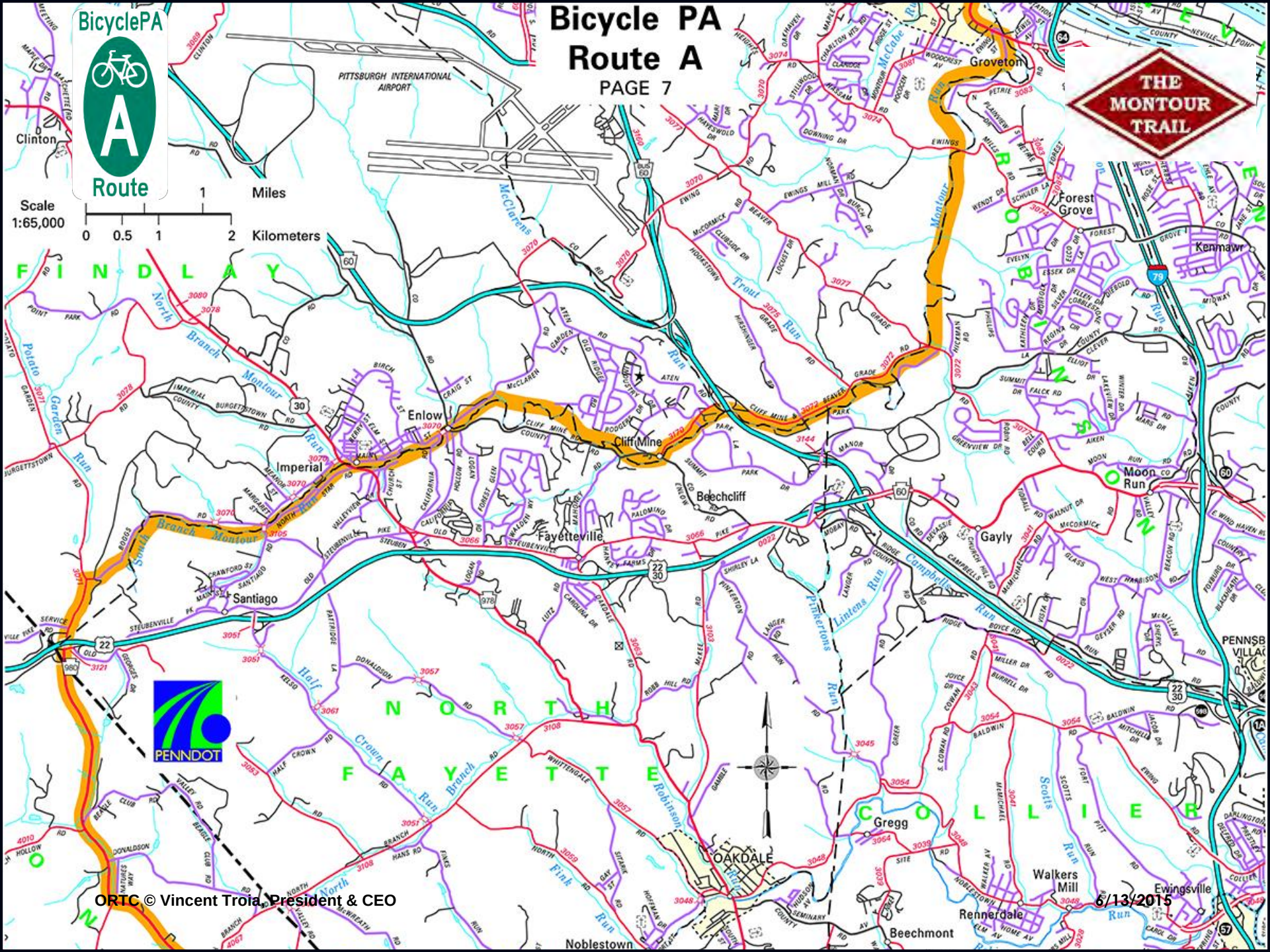
Route

Scale
1:65,000

0 0.5 1 2 Miles
0 0.5 1 2 Kilometers

Bicycle PA Route A

PAGE 7



High Speed Environment
PA Route 51
PA Bike Route A
Ambridge Bridge



High Speed Environment
PA Route 51 - PA Bike Route A
Hopewell Township, Pa



High Speed Environment
PA Route 51 - PA Bike Route A
South Heights, Pa



High Speed Environment
PA Route 51 - PA Bike Route A
Stoops Ferry - Moon Township, Pa



PA Route 51 - PA Bike Route A
Moon Township, Pa



6/13/2015

Cycling Safety – PA 51 Bike Route A



Fatalities

- ▶ The ORTC is tasked to develop a **safe bicycle path** along the Ohio River from Coraopolis and Glen Osborne to the PA–Ohio State line, which includes improving **Bike PA Route A**.



**Cyclist Fatality
Phase 2 Location**

Route 51

**Cyclist Fatality
Phase 1 Location**

BicyclePA



A

Route

Cycling Safety – PA Bike Route A



Emily Rose Jancart

- ▶ Age 17, of Moon Township, a recent graduate of Our Lady of Sacred Heart High School and accepted at Carlow University as a nursing major, was lifted up in the arms of God from a tragic bicycle accident around 5:30 p.m. on Sunday, July 21, 2013 in Moon Township, Pa.



Emilie Rose Jancart
Ghost Bike



Emilie Rose Jancart
Ghost Bike



6/13/2015

Cyclist Fatalities – PA Bike Route A



Taylor Lee Banks

- ▶ Age 23, of Braddock, was tragically struck by a vehicle just after 7:00 p.m. near the West Aliquippa Bridge on October 31, 2014 while bicycling home on Route 51 from his workplace at the Subway in Aliquippa to his mother's house in Rochester, Pa.

**Taylor Lee Banks
Ghost Bike**



Bike PA Route A - Traffic Fatalities



Vision Zero

- ▶ A road traffic safety project started in Sweden in 1997 which aims to achieve a highway system with no fatalities or serious injuries in road traffic.
- ▶ Boston, New York, and San Francisco introduced initiatives in 2014.

Bike PA Route A - Traffic Fatalities



45

Vision Zero

- ▶ Mayor William Peduto Issued an Executive Order On **Complete Streets Policy** for Pittsburgh Rights of Way on April 10, 2015.
- ▶ Included the “**vision of zero traffic fatalities.**”



Bike PA Route A - Traffic Fatalities



Vision Zero

- ▶ “The complete streets policy and framework shall focus on safety and accessibility for people of all ages and abilities, a strategy and implementation plan for robust data collection and analysis, context-appropriate design guidelines, and a plan for ongoing education and enhanced enforcement measures to achieve the **vision of zero traffic fatalities.**”

Bike PA Route A - Traffic Fatalities



Vision Zero

- ▶ The Ohio River Trail Council has adopted a “Vision Zero” plan to eliminate traffic fatalities and serious injuries along the Ohio River Trail corridor by 2025.



Bicycle Path Classifications

Dr. Vincent Troia, CEO



Bicycle Path – On-Road



Classifications

- ▶ Signed – Class III
- ▶ Shared – Class III
- ▶ Designated – Class II
- ▶ Separated – Class I
(most safest)





Bicycle Path

On-Road Class III

Signed Right-of-Way

Bicycle Path – On-Road Class III



Signed





Bicycle Path – On-Road Class III



Signed

- ▶ Provides wayfinding guidance.
- ▶ A preferred Bicycle Route.
- ▶ No pavement markings.





Bicycle Path

On-Road Class III

Shared Right-of-Way

Bicycle Path – On-Road Class III



55

Signed & Shared

► Bike Lanes Types

- ❖ **Share-the-Road Signs**
- ❖ Alerts drivers that bicyclists are likely to be operating in traffic.



Bicycle Path – On-Road Class III



Shared

- ▶ Bike Lanes Types
 - ❖ **Shared Pavement Markings**
 - ❖ Adds pavement markings





Bicycle Path

On-Road Class II

Designated Right-of-Way

Bicycle Path – On-Road Class II



Designated

- ▶ Bike Lanes Types
 - ❖ **Shoulder**
 - ❖ **Conventional**
- ▶ Adds pavement striping



Bicycle Path – On-Road Class II



59

Designated

Shoulder Bike Lane



Conventional Bike Lane



Bicycle Path – On-Road Class II



60

Designated

► Bike Lanes Types

❖ **Conventional Painted**



Bicycle Path – On-Road Class II



Designated

► Bike Lane Types

- ❖ **Buffered** - Adds a **painted buffer** to increase lateral separation between bicyclists and motor vehicles.



Bicycle Path – On-Road Class II



Designated

► Bike Lane Types

- ❖ **Buffered** - Adds a **painted buffer** to increase lateral separation between bicyclists and parked cars.



Bicycle Path – On-Road Class II



Designated

► Bike Lane Types

❖ Buffered



Bicycle Path – On-Road Class II



Designated

► Bike Lane Types

- ❖ Painted two-way bike lane with buffer.



Bicycle Path – On-Road Class II

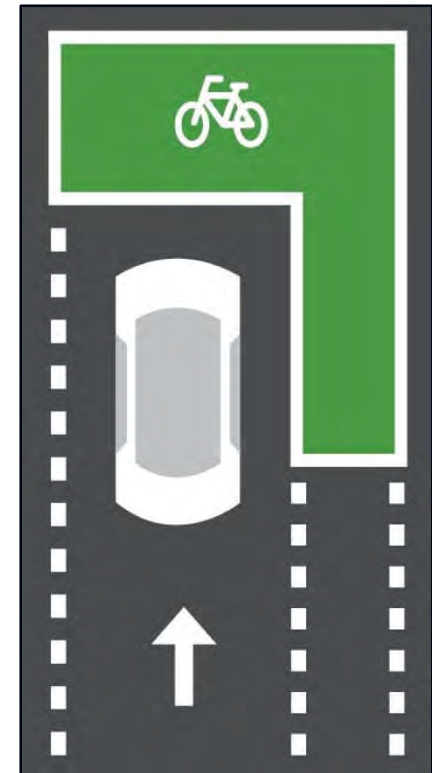


65

Designated

► Bike Lane Types

- ❖ **Bike Box** – A painted green space on the road with a white bicycle symbol inside.



Bicycle Path – On-Road Class II



66

Designated

► Bike Lane Types

- ❖ **Bike Box** – Permits bikes to enter the intersection **ahead of traffic.**



Bicycle Path – On-Road Class II



Designated

► Bike Lane Types

- ❖ **Bike Box** – Makes bikes **more visible** to drivers.

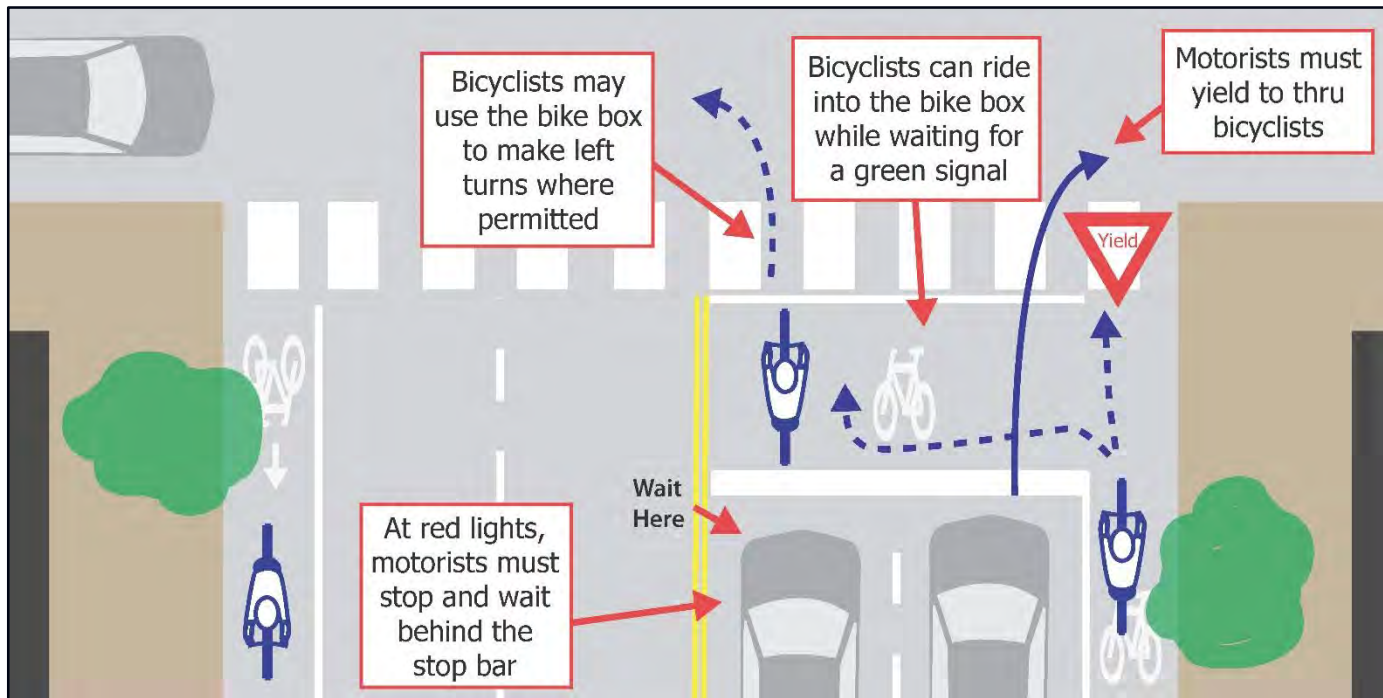


Bicycle Path – On-Road Class II



68

Designated (Bike Box)



Bicycle Path – On-Road Class II



Designated

► Bike Lane Types

❖ Intersection

- ❖ **45%** of bicycle/motor vehicle collisions occur at intersections

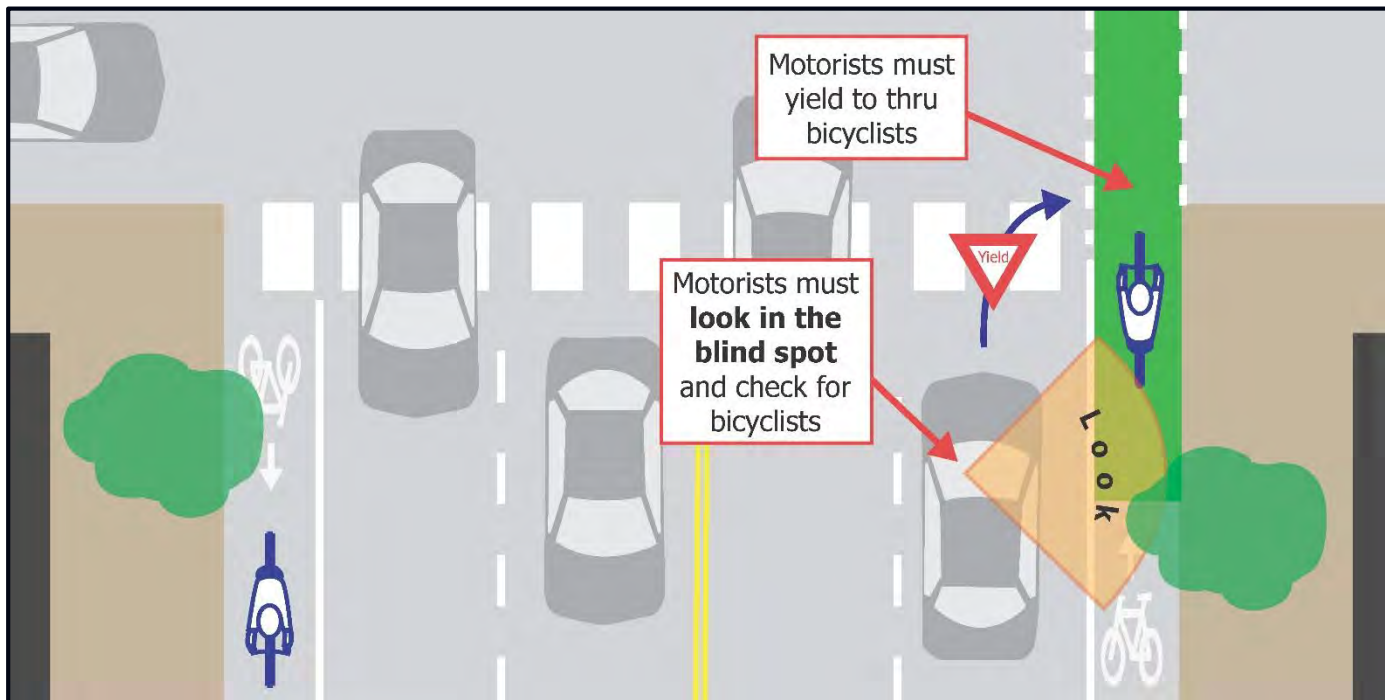


Bicycle Path – On-Road Class II



70

Designated (Intersection)



Bicycle Path – On-Road Class II



Designated

- ▶ Bike Lane Types
 - ❖ Intersection





Intersection Bike Lane
Penn Ave, Pittsburgh

Intersection Bike Lane



**Bike Lane to Share-the-Road Transition
Bigelow Blvd, Oakland**

Bicycle Path – On-Road Class II

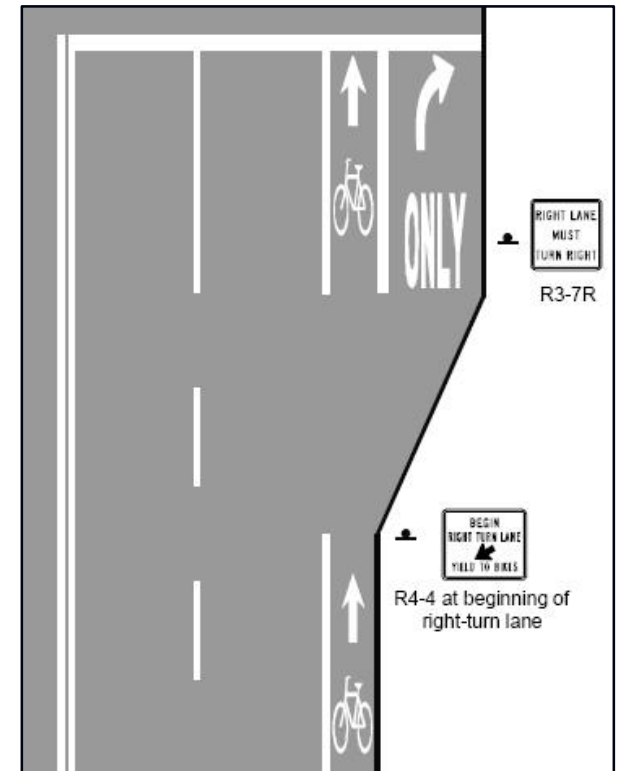


75

Designated

► Bike Lane Types

- ❖ **Pocket Bike Lane** –
A section of bike lane that has a lane for vehicles on either side.





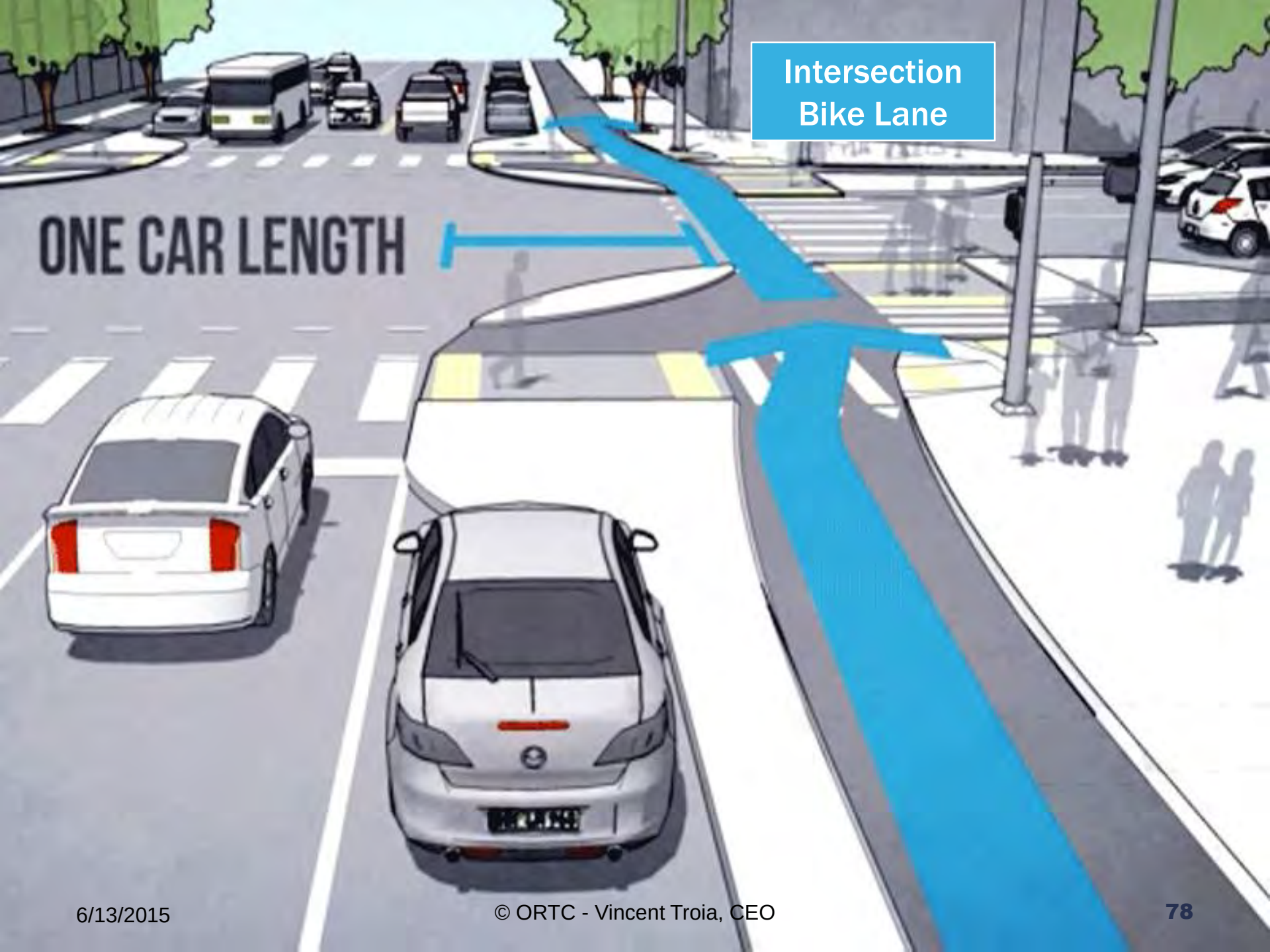
Pocket Bike Lane



Pocket Bike Lane
Bigelow Blvd, Oakland

Intersection Bike Lane

ONE CAR LENGTH





Bicycle Path

On-Road Class I

Separated Right-of-Way

Bicycle Path – On-Road Class I



80

Separated

► Protected Bike Lane

- ❖ **Separated** from motor vehicle traffic with a physical barrier or **vertical element**.
- ❖ Exclusive facility for bicyclists.

Bicycle Path – On-Road Class I



81

Separated

► Protected Bike Lane Types

- ❖ Delineator or Bollard
- ❖ Planter
- ❖ Curb
- ❖ Parked Cars
- ❖ Raised

Bicycle Path – On-Road Class I



82

Separated

► Protected Bike Lane

❖ Delineator





Protected Bike Lane with Delineators
Sixth Street, Pittsburgh

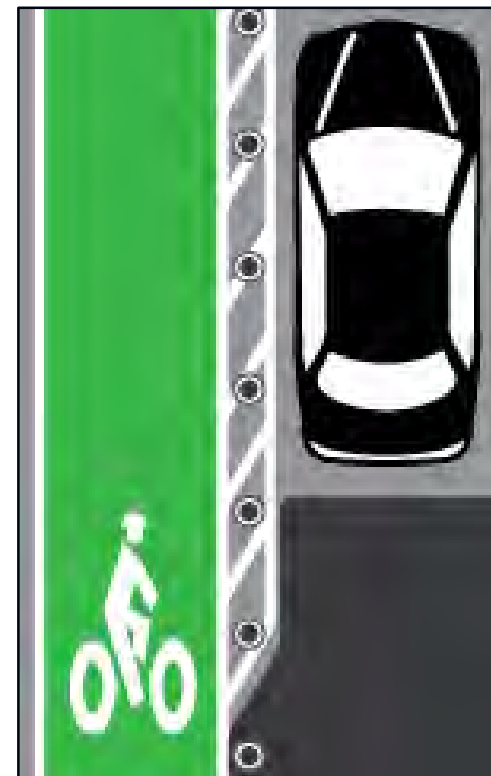
Bicycle Path – On-Road Class I



Separated

► Protected Bike Lane

- ❖ Painted bike lane that is buffered and protected with a delineator



**Two-Way Protected Buffered
Bike Lane with Delineators
Schenley Park, Pittsburgh**



Two-Way Protected Buffered Bike Lane with Delineators Schenley Park, Pittsburgh

86





**Two Way Protected Buffered Bike Lane
with Delineators - Schenley Park, Pittsburgh**

Bicycle Path – On-Road Class I



88

Separated

► Protected Bike Lane

❖ Planter





89

Bicycle Path – On-Road Class I



90

Separated

► Protected Bike Lane

❖ Curb



Bicycle Path – On-Road Class I

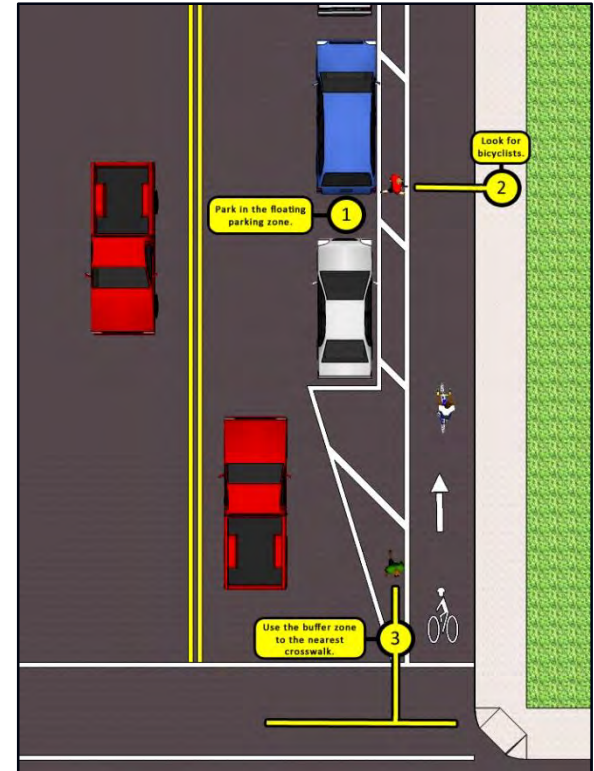


91

Separated

► Protected Bike Lane

❖ Floating Parking Lane



Protected Bike Lane



Parked Cars

Protected Bike Lane

Cars

Delineator

Planter

Buffer Zone



Bicycle Path – On-Road Class I

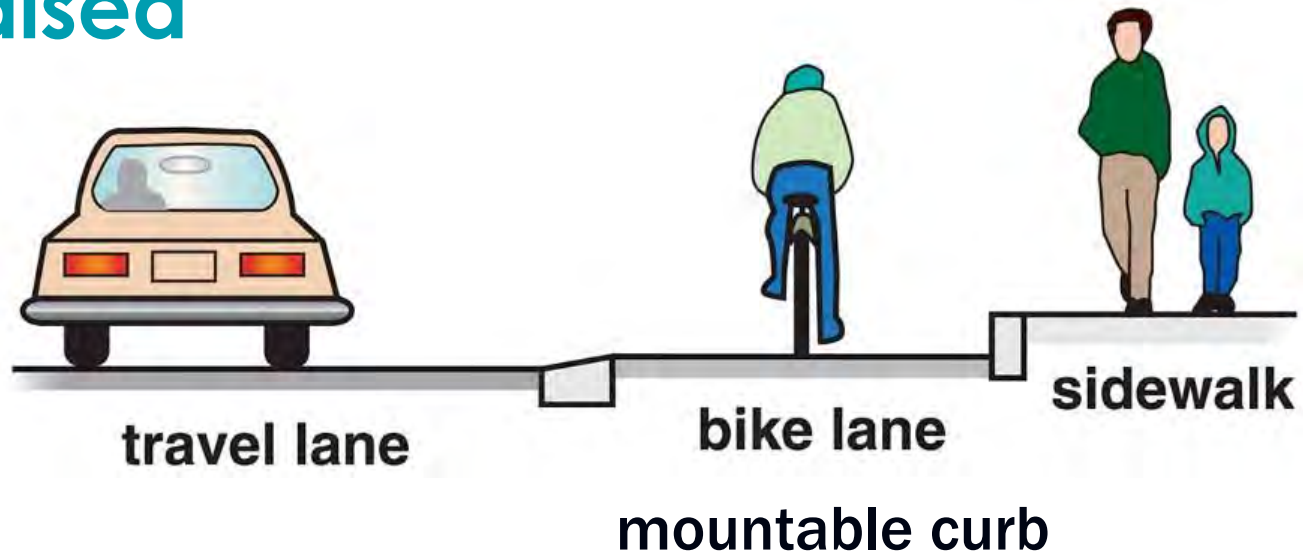


94

Separated

► Protected Bike Lane

❖ Raised



Bicycle Path – On-Road Class I



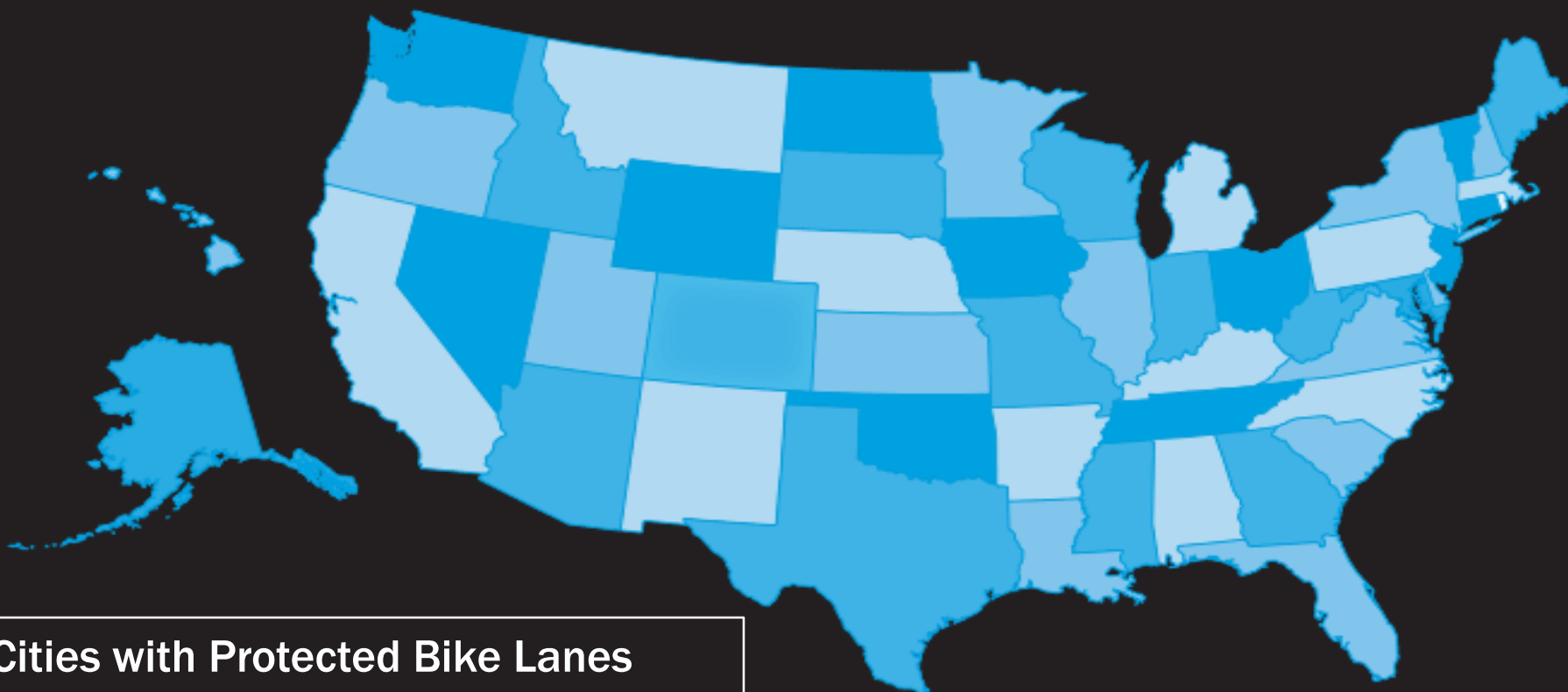
95

Separated

► Protected Raised Bike Lane with curb



GROWTH OF THE PROTECTED BIKE LANE



**Cities with Protected Bike Lanes
grew from 16 to 60 from 2002-2015**



peopleforbikes™

Number of U.S protected bike lanes:

Number of Protected Bike Lanes increased
from 27 in 2002 to 196 in 2015

2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015



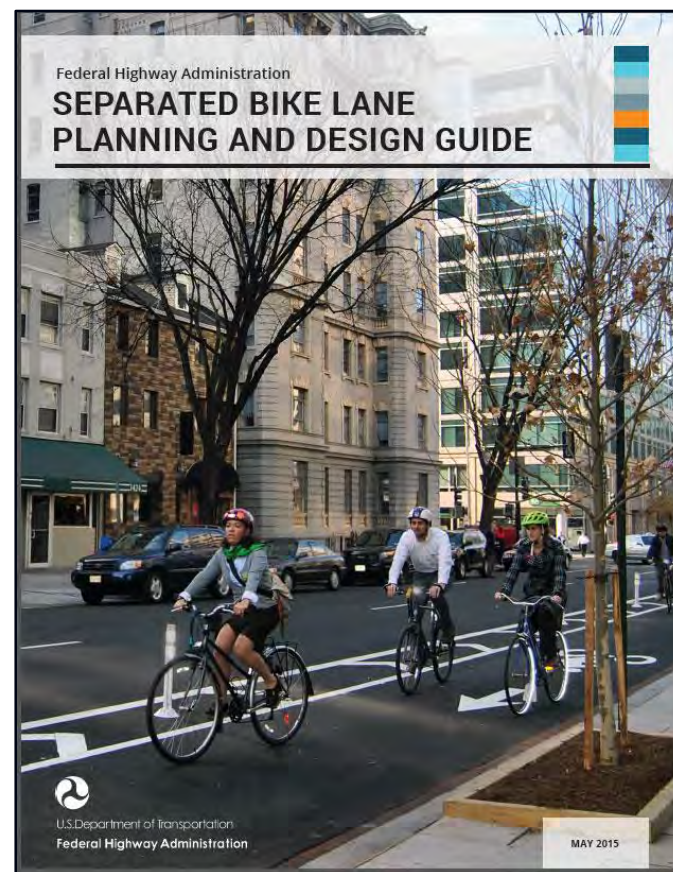
peopleforbikes™

Bike Path – On-Road Class I



Growth of Protected Bike Lanes

- ▶ Protected Bike Lanes – Official Federal Highway Administration Policy.
- ▶ “Separated Bike Lane Planning and Design Guide” – May 2015.



Bike Path – On-Road Class I



99

Growth of Protected Bike Lanes

- ▶ “Protected Bike Lanes are the **next-generation** bikeways being built across the U.S.”

– People for Bikes



Bicycle Path – On-Road Class I



100

The first Separated Bike Path?

- ▶ Coney Island Cycle Path, June 15, 1894
- ▶ Prospect Park, Brooklyn to Coney Island.
- ▶ “The first path in the world devoted exclusively to bicycles,”
– Brooklyn Daily Eagle.

**Coney Island Cycle Path
Ocean Parkway
Built in 1894**

Bicycle Path – On-Road Class I



Coney Island Cycle Path

- ▶ Brooklyn's 80,000 cyclists championed the path's completion due to their love affair with the bicycle.
- ▶ Initially 5.5 miles long and 14 feet in width.
- ▶ It was widened by three feet along its entire length in December and then widened again to 34 feet in 1896.

**Coney Island Cycle Path
Prospect Park, Brooklyn to Coney Island
Gala Parade 1896**



Bicycle Path – On-Road Class I



Coney Island Cycle Path

- ▶ In June 1896 a return path was built on the opposite side of the boulevard.
- ▶ Opened with a gala parade organized by the League of American Wheelmen's Good Roads Association
- ▶ Attended by 10,000 cyclists from 60 cycling clubs and upwards of 100,000 spectators.

The oldest greenway in the U.S. is the Ocean Parkway Bike Path in Brooklyn.



**Ocean Parkway Bike Path
Carroll Gardens, Brooklyn
to Coney Island**



Franklin D. Roosevelt
Boardwalk and Beach
6/13/2015



Bicycle Path

Off-Road Class I

Separated Right-of-Way

Bicycle Path – Off-Road



108

Classifications

- ▶ Separated – Class I (most safest)
 - ❖ Urban
 - ❖ Rural





Bicycle Path Class I Separated Right-of-Way Urban Off-Road





Bicycle Path Class I Separated Right-of-Way Rural Off-Road



Bicycle Path – Off-Road Class I



111

Separated

- ▶ *Multi-use Trails - Intended for use by a variety of groups, including hikers, runners, and cyclists.*



Bicycle Path – Off-Road

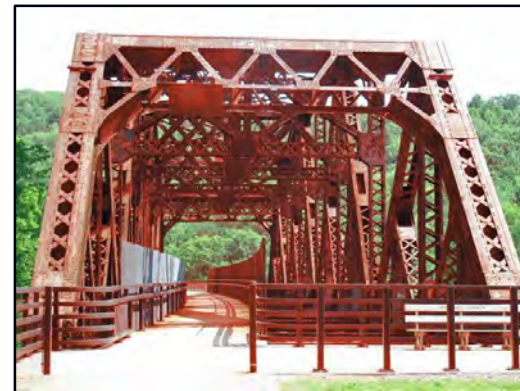
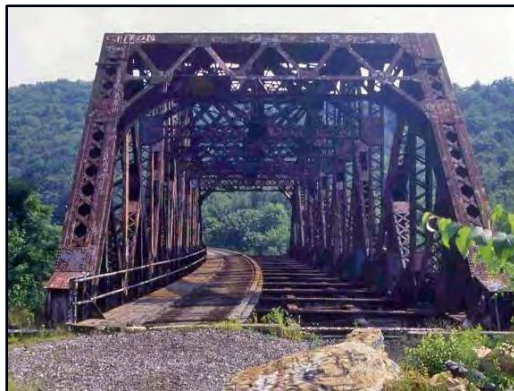


112

Separated

► Multi-use Trails (Rails-to-Trails)

GAP Keystone Viaduct-Meyersdale, Pa



Bicycle Path – Off-Road



113

Separated

- ▶ *Multi-use Trails (Rails-with-Trails)*
 - ❖ *Along an active railroad*
 - ❖ *Little Beaver Creek Greenway Trail*



Bicycle Path – Off-Road Class I



114

Separated

► Multi-use Trail Road Crossing

- ❖ **Safety** Concerns

- ❖ Stop Signs



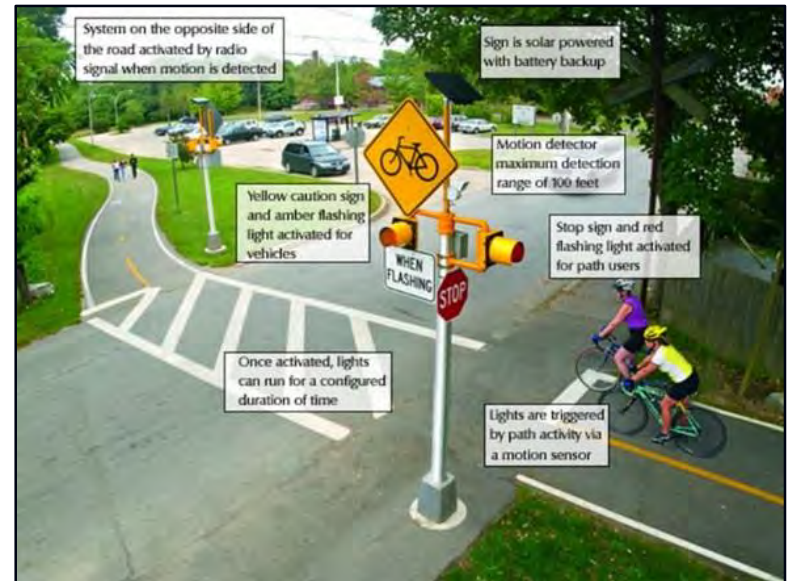
Bicycle Path – Off-Road Class I



115

Separated

- ▶ Trail Road Crossing
 - ❖ **Safety** Concerns
 - ❖ Cross Alert **Signal** Warning System



Bicycle Path – Off-Road Class I



116

Separated

- ▶ Trail Road Crossing
 - ❖ Montour Trail
 - ❖ Cliff Mine Road
 - ❖ Cross Alert Installed
 - ❖ November 15, 2008



Ohio River Greenway Trail

Phase 1 & 2 Studies

John Orndorff, Director



Ohio River Trail Phase 1 & 2 Study



118

John Orndorff

- ▶ *John Orndorff, Director*
- ▶ *Ohio River Trail Council*
- ▶ *ORGT Phase 2 Chairman*
- ▶ *Glen Osborne Council, Vice President*
- ▶ *Email: orndjo@gmail.com*

Ohio River Trail Phase 1 & 2 Study



119

Ohio River Trail Council

- Incorporated May 2009 as a non-profit corporation to establish a multi-use trail along the Ohio River from the Ohio state line to the Montour Trail in Coraopolis.



Ohio River Trail Phase 1 & 2 Study



120

Ohio River Trail Council

- ▶ To provide recreation, transportation, and the conservation of natural and historic resources.
- ▶ To promote economic benefits to the River & Trail Towns.

Ohio River Trail Phase 1 & 2



121

Ohio River Trail Council

- ▶ The ORTC relies on corporate, foundation, government and private **donations** to achieve its mission.



Ohio River Trail Phase 1 & 2 Study



122

Ohio River Trail Council

- ▶ **Bicycle Friendly Communities**
- ▶ To foster opportunities for healthier life styles and enhance the quality of life for residents.



Ohio River Trail Phase 1 & 2 Study



123

Study Objectives

- ▶ Improve **safety** for cyclists and motorists.
 - ❖ Identify location and types of signs and pavement markings.
 - ❖ Recommend improvements to intersections and shoulders.

Ohio River Trail Phase 1 & 2 Study



124

Study Objectives

- ▶ Estimate costs for materials and installation.
- ▶ Submit a report for PennDOT consideration.



Ohio River Trail Phase 1 & 2 Study



125

Study Corridor

- ▶ Phase 2 - Center Township (Beaver County Jail) to Moon Township.
- ▶ Phase 1 – Moon Township to Coraopolis.

Ohio River Trail Phase 1 & 2 Study



126

Phase 2 Communities

- ▶ Center Township
- ▶ Aliquippa
- ▶ Hopewell
- ▶ South Heights
- ▶ Ambridge
- ▶ Baden
- ▶ Leetsdale
- ▶ Edgeworth
- ▶ Sewickley
- ▶ Glen Osborne

Ohio River Trail Phase 1 & 2 Study



127

Phase 1 Communities

- ▶ Coraopolis
 - ❖ Ray McCutcheon, Manager
 - ❖ manager@coraopolispa.com
- ▶ Moon Township
 - ❖ Lance Welliver, Director Moon Parks
 - ❖ llwelliver@moonparks.org



BEAVER COUNTY JAIL

Ohio River Greenway Trail
Beaver County Jail Trail
Center Township
Phase 2 Trailhead



8/13/2015

Ambridge



Ohio River Greenway Trail Phase 2 Corridor

Leetsdale

Edgeworth

Sewickley

Glen Osborne



**Ohio River Greenway Trail
Phase 1 Corridor**



**Phase 1 connects to the planned
Moon Township Waterfront park**

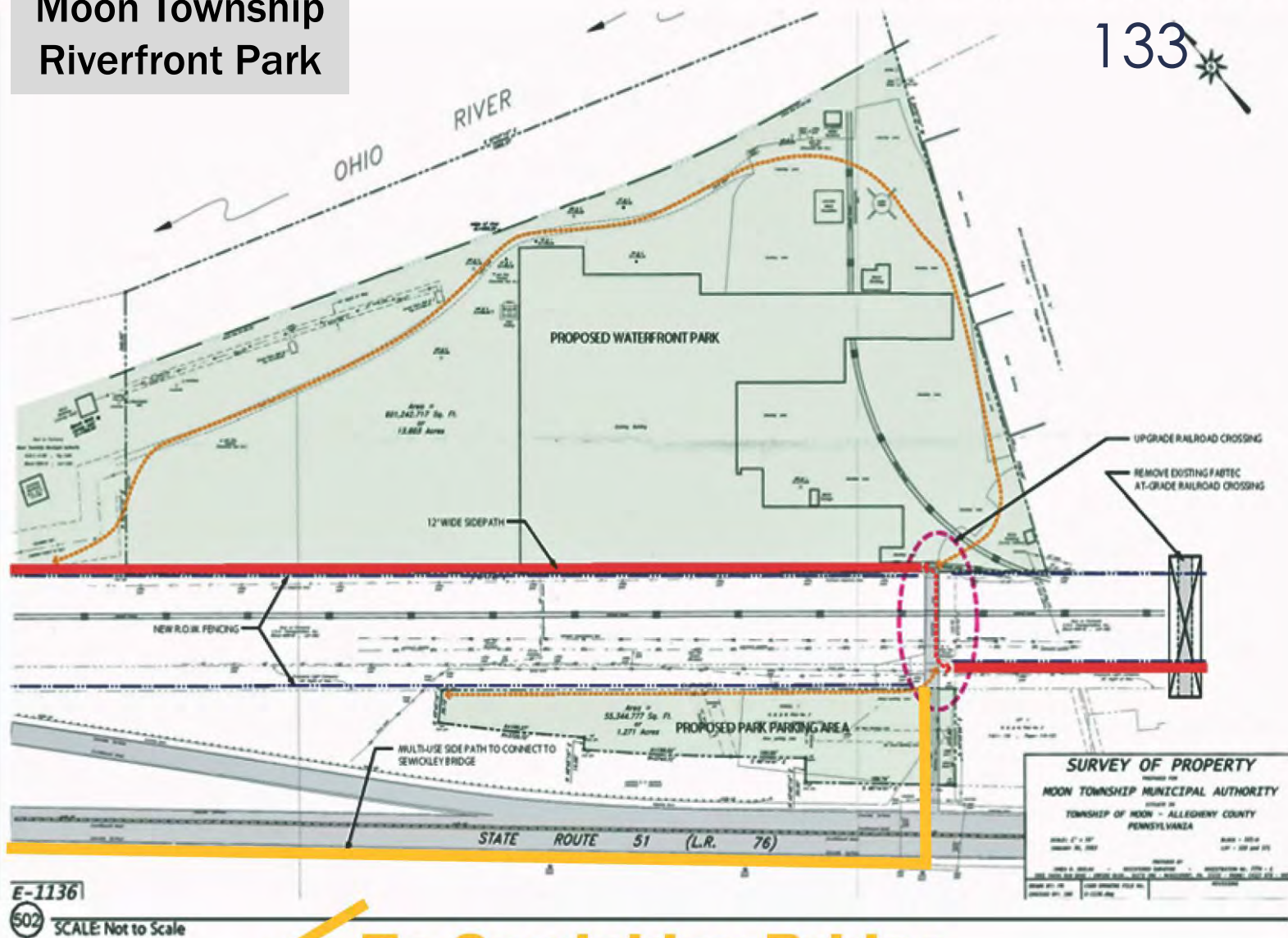
LEGEND

- ◆ Signed On-Road Route
- ◆ Neighborhood Feeder Route
- Off-Road Route
- Trail Blazer Sign

Ohio River Trail Moon Township Riverfront Park

Moon Township Municipal Authority Property - 502

133



E-1136

502 SCALE: Not to Scale

← To Sewickley Bridge

Ohio River Trail – Phase 1



Moon Township Waterfront Park

- ▶ Multi-use Trailhead
 - ❖ Ohio River Greenway Trail
 - ❖ Ohio River Water Trail
 - ❖ Three Rivers Water Trail



Allegheny County Sports & Athletic Complex at Montour Junction



Ohio River Trail Phase 1 Southern Terminus

-
- PHASE II**
- (6) ADDITIONAL TURF SPORTS FIELDS
 - PICNIC PAVILIONS AND ADDITIONAL PARKING
 - ADDITIONAL FITNESS TRAILS
- PHASE I**
- EMERGENCY ACCESS/OVERFLOW PARKING
 - COMMUNITY PLAYGROUND AREA
 - BRIDGE ACROSS MONTOUR RUN
-PEDESTRIAN & VEHICULAR ACCESS
 - FEATURE FIELD
-SYNTHETIC FIELD
-SEATING
-TRAINING AND LOCKER ROOMS
-CONCESSION/RESTROOM SPACE
 - BIKE/FITNESS TRAIL
 - CONNECTION TO MONTOUR TRAILHEAD
- PHASE II**
- ADDITIONAL PARKING
- PHASE I**
- (3) SYNTHETIC PRACTICE FIELDS
- OVERFLOW PARKING
- PHASE I BOUNDARY

MASTER PLAN - WHOLE SITE
07.15.2010



BURT, HILL

SCALE: 1"=100'



136



Mackin Engineering

Ohio River Greenway Trail Phase 1 & 2

Bob Genter, Director



Mackin Engineering



Bob Genter

- ▶ *Robert Genter, Director*
- ▶ *Mackin Engineering Company*
- ▶ *p: 412.788.0472 | c: 412.496.9210*
- ▶ *rwg@mackinengineering.com*



Mackin Engineering



Chuck Jones

- ▶ *Chuck Jones*
- ▶ *Traffic Engineer*
- ▶ *Mackin Engineering Company*
- ▶ *p: 412.788.0472 | c: 412.496.9210*



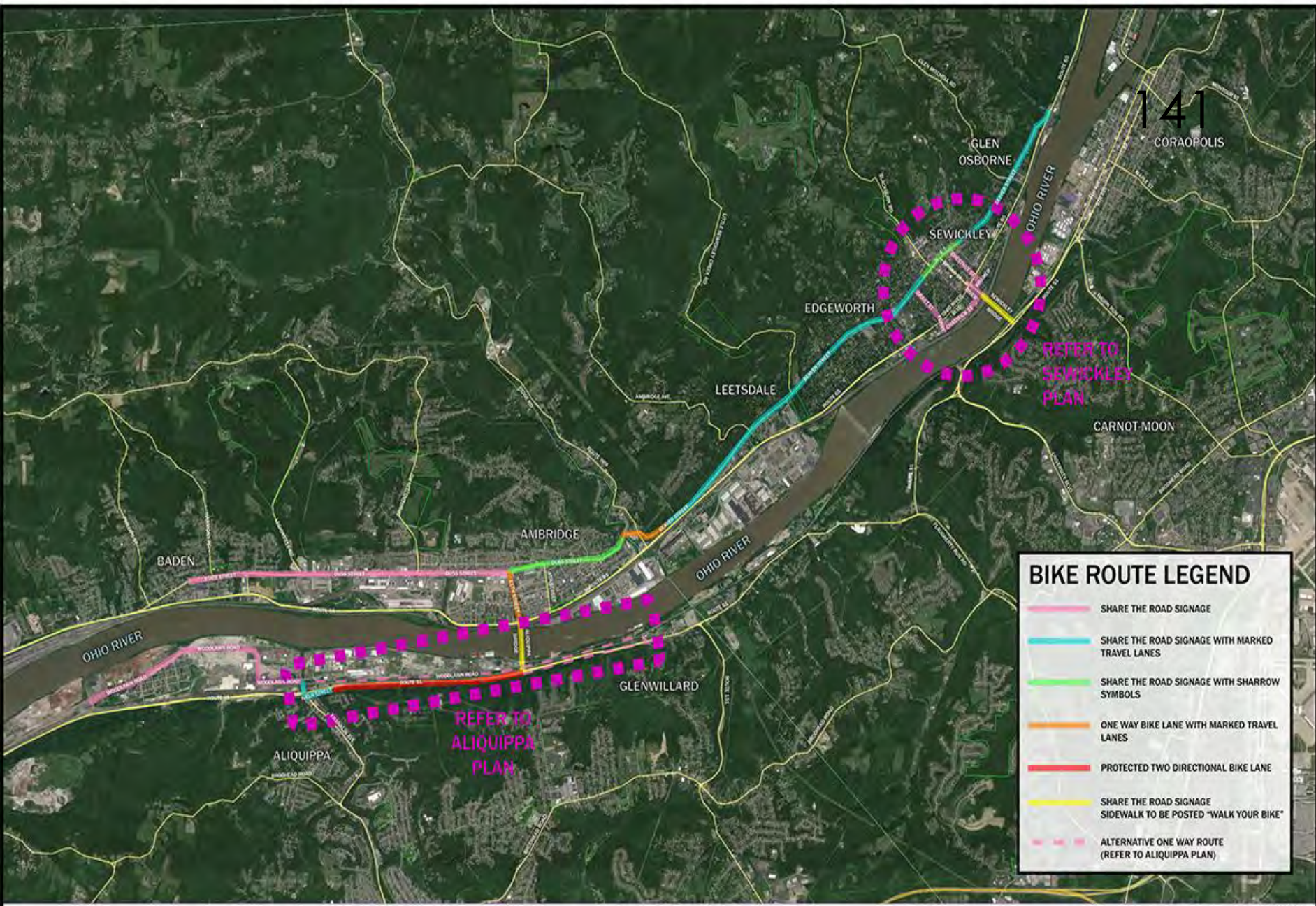


Ohio River Greenway Trail

Mackin Engineering

Phase 2

Center Twp. to Moon Twp.



BIKE ROUTE LEGEND

- SHARE THE ROAD SIGNAGE
- SHARE THE ROAD SIGNAGE WITH MARKED TRAVEL LANES
- SHARE THE ROAD SIGNAGE WITH SHARROW SYMBOLS
- ONE WAY BIKE LANE WITH MARKED TRAVEL LANES
- PROTECTED TWO DIRECTIONAL BIKE LANE
- SHARE THE ROAD SIGNAGE SIDEWALK TO BE POSTED "WALK YOUR BIKE"
- ALTERNATIVE ONE WAY ROUTE (REFER TO ALIQUIPPA PLAN)

**OHIO RIVER TRAIL ON ROAD CONNECTOR PLAN
OVERALL CONCEPT PLAN**



**OHIO RIVER TRAIL ON ROAD CONNECTOR, CITY OF ALIQUIPPA
OVERALL CITY PLAN**

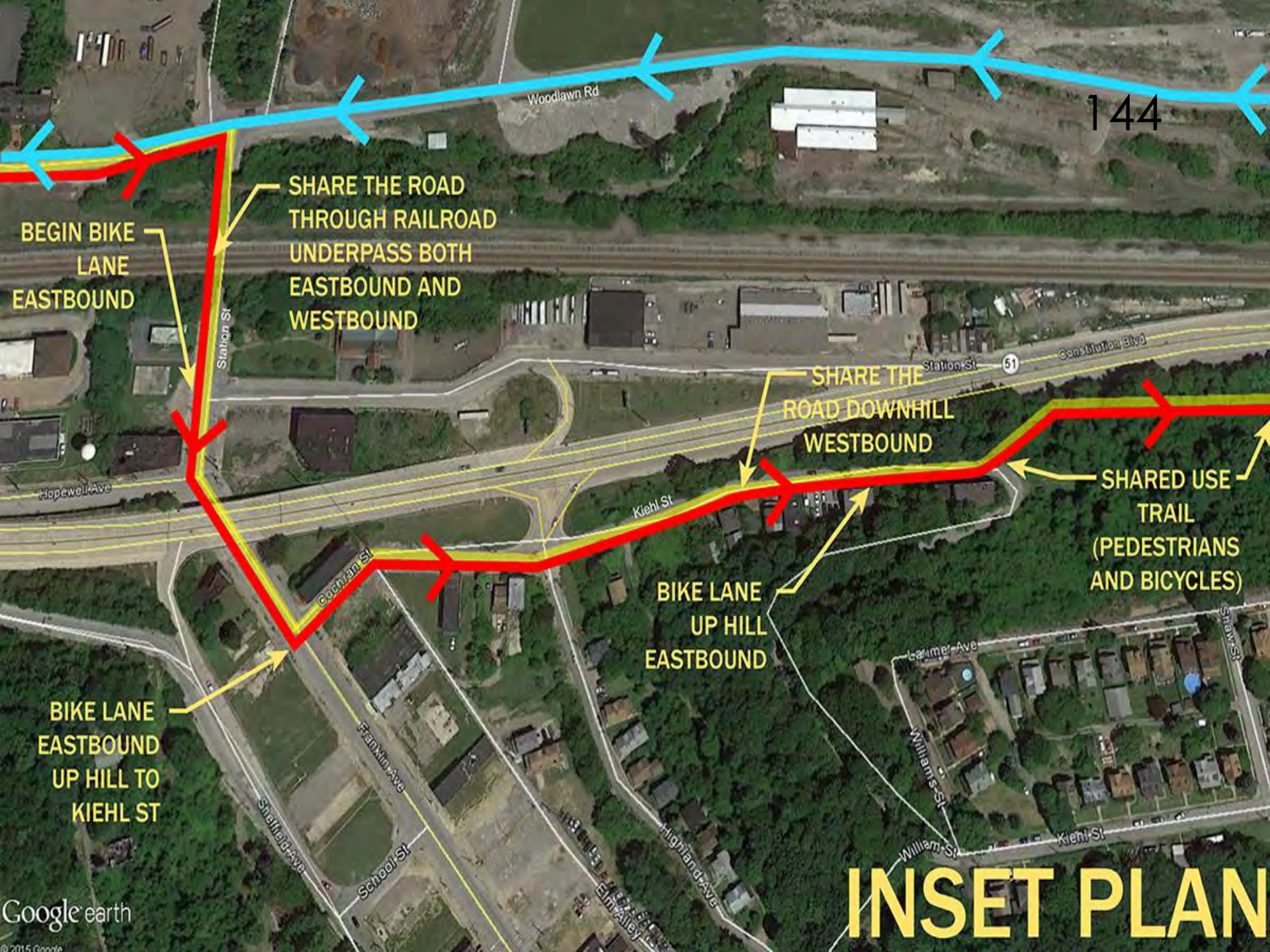


Ohio River Greenway Trail

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Phase 2

Aliquippa



144

BEGIN BIKE LANE
LANE
EASTBOUND

SHARE THE ROAD
THROUGH RAILROAD
UNDERPASS BOTH
EASTBOUND AND
WESTBOUND

SHARE THE
ROAD DOWNHILL
WESTBOUND

SHARED USE
TRAIL
(PEDESTRIANS
AND BICYCLES)

BIKE LANE
UP HILL
EASTBOUND

BIKE LANE
EASTBOUND
UP HILL TO
KIEHL ST

INSET PLAN

EXISTING CONDITIONS



EXISTING CURB TO
CURB WIDTH = 25'

SHARE THE ROAD
SIGNAGE



BIKE LANE
'UPHILL'
(EASTBOUND)



SHARED LANE MARKING
THROUGH INTERSECTION
(WESTBOUND)

PROPOSED BIKE LANE = x1 @ 5'
PROPOSED DRIVE AISLES = x2 @ 10'

OHIO RIVER TRAIL ON ROAD CONNECTOR, CITY OF ALIQUIPPA
KIEHL STREET MODIFICATION PLAN

APRIL 2015
Mackin

146

Image Landsat

Google earth



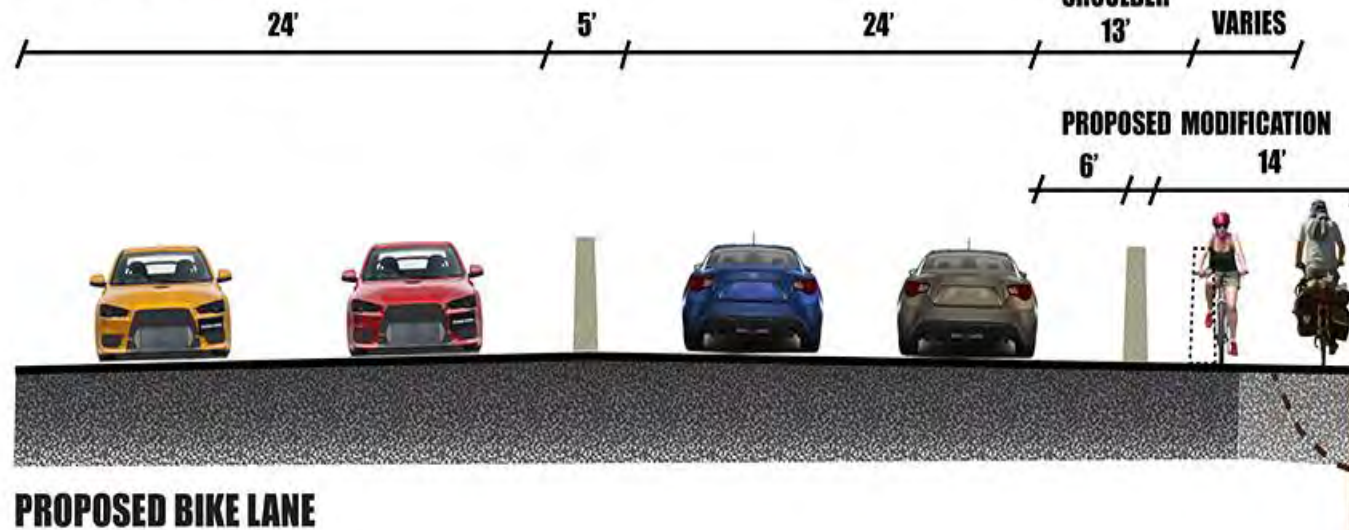
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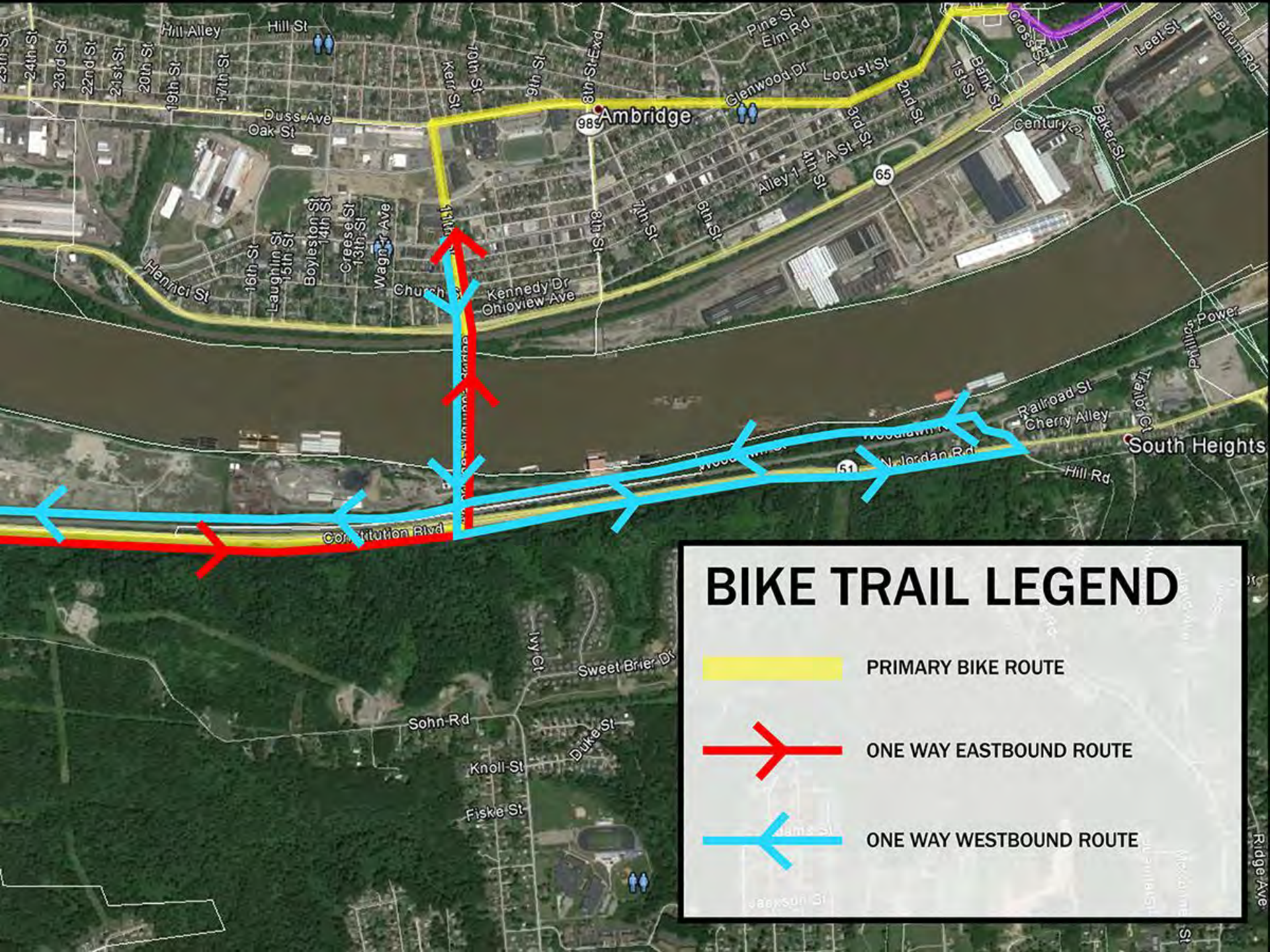
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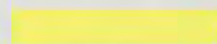
EXISTING CONDITIONS



OHIO RIVER TRAIL ON ROAD CONNECTOR ROUTE 51 MODIFICATION PLAN



BIKE TRAIL LEGEND



PRIMARY BIKE ROUTE



ONE WAY EASTBOUND ROUTE



ONE WAY WESTBOUND ROUTE



Ohio River Greenway Trail

Mackin Engineering

Phase 2 Ambridge

EXISTING CONDITIONS



EXISTING CURB TO CURB WIDTH = 30'

152

SHARE THE ROAD
SIGNAGE



DEDICATED BIKE LANE
WITH PAINTED SYMBOL



PROPOSED BIKE LANE = x1 @ 6'
PROPOSED DRIVE AISLES = x2 @ 11'
PROPOSED SHOULDER = x1 @ 2'

OHIO RIVER TRAIL ON ROAD CONNECTOR, BOROUGH OF AMBRIDGE
11TH STREET MODIFICATION PLAN

APRIL 2015

Mackin

EXISTING CONDITIONS

153

EXISTING CURB TO
CURB WIDTH = 30'

SHARE THE ROAD
SIGNAGE

DEDICATED BIKE LANE
WITH PAINTED SYMBOL

PROPOSED BIKE LANE = x1 @ 6'
PROPOSED DRIVE AISLES = x2 @ 11'
PROPOSED SHOULDER = x1 @ 2'

**OHIO RIVER TRAIL ON ROAD CONNECTOR, BOROUGH OF AMBRIDGE
11TH STREET MODIFICATION PLAN**

EXISTING CONDITIONS



EXISTING CURB TO
CURB WIDTH = 35'

**SHARROWS; INSTALL IN
BOTH TRAVEL LANES**

**SHARE THE ROAD
SIGNAGE**

**PAINTED PARALLEL
PARKING STALLS**

154

OPTION A

EXISTING DRIVE AISLES = x2 @ 14'
EXISTING PARALLEL PARKING ROW = x1 @ 7'

**DEDICATED BIKE LANE
WITH PAINTED SYMBOL**

**SHARE THE ROAD
SIGNAGE**

OPTION B

PROPOSED DRIVE AISLES = x2 @ 12.5'
PROPOSED BIKE LANES = x2 @ 5'
(LOSS OF PARALLEL PARKING)

**OHIO RIVER TRAIL ON ROAD CONNECTOR, BOROUGH OF AMBRIDGE
DUSS AVENUE MODIFICATION PLAN**

EXISTING CONDITIONS

EXISTING CURB TO
CURB WIDTH = 34'

155

SHARE THE ROAD
SIGNAGE

ROAD EDGE
MARKING

PROPOSED DRIVE AISLES = x2 @ 11'
PROPOSED SHOULDERS = x2 @ 6'

**OHIO RIVER TRAIL ON ROAD CONNECTOR, BOROUGH OF AMBRIDGE
BEAVER STREET MODIFICATION PLAN**

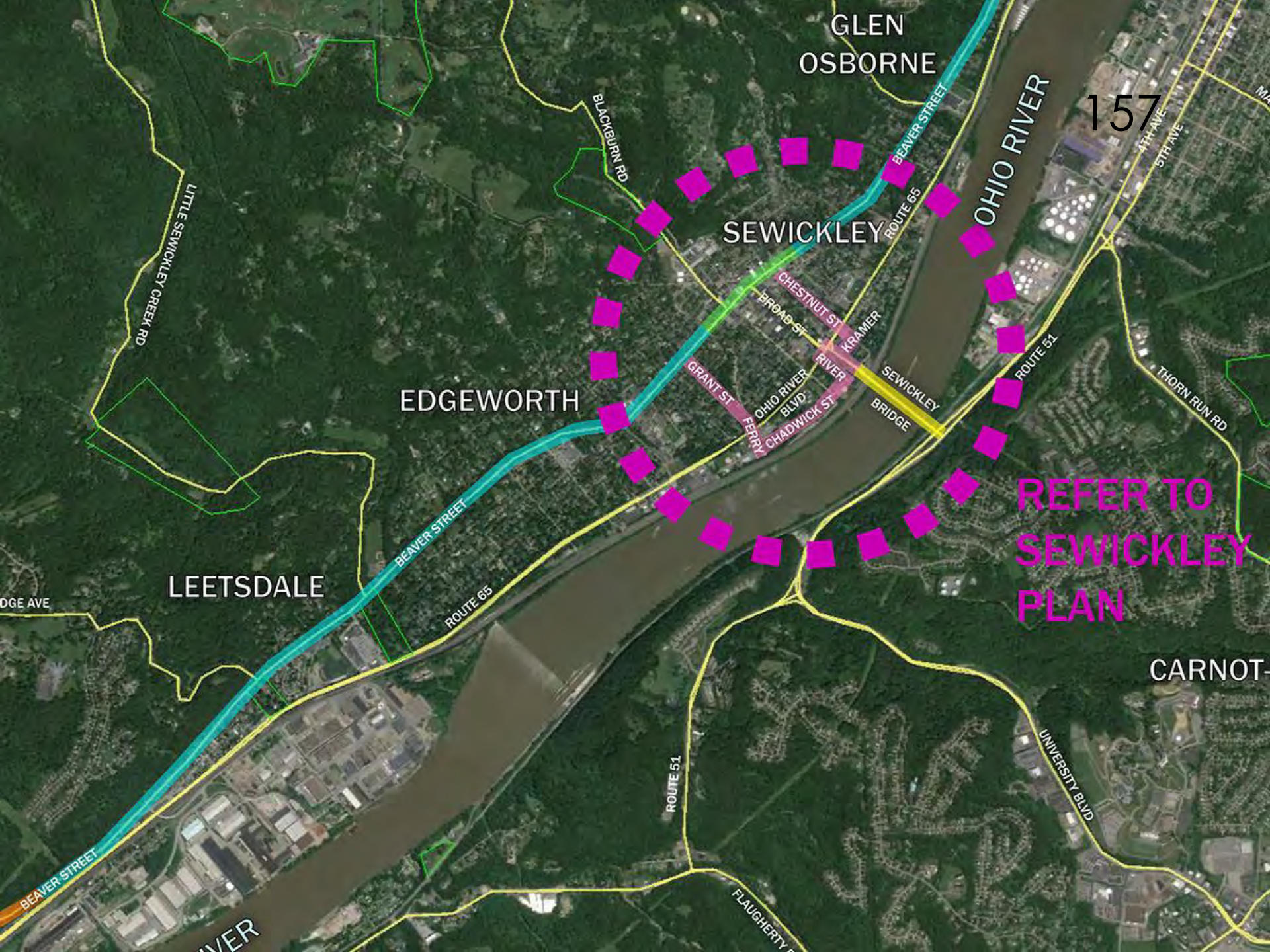


Ohio River Greenway Trail

Mackin Engineering

Phase 2

Leetsdale



GLEN
OSBORNE

157

SEWICKLEY

EDGEWORTH

LEETSDALE

REFER TO
SEWICKLEY
PLAN

CARNOT-

EXISTING CONDITIONS

EXISTING SIDEWALK
WIDTH = 5'

EXISTING CURB TO
CURB WIDTH = 25'

158

SAWCUT AND REINSTALL
EXTENDED WIDTH SIDEWALK
FOR DEDICATED BIKE LANE

SHARE THE ROAD
SIGNAGE

PROPOSED BIKE LANE = x1 @ 8'
PROPOSED DRIVE AISLES = x2 @ 11'
PROPOSED SHOULDER = x1 @ 1'

**OHIO RIVER TRAIL ON ROAD CONNECTOR, BOROUGH OF LEETSDALE
BEAVER STREET MODIFICATION PLAN**



Ohio River Greenway Trail

Mackin Engineering

Phase 2

Edgeworth

EXISTING CONDITIONS

EXISTING CURB TO
CURB WIDTH = 30'-3"

160

SHARE THE ROAD
SIGNAGE



EXISTING ROAD
EDGE MARKING

EXISTING DRIVE AISLES = x2 @ 11
EXISTING SHOULDERS = x2 @ 4.17

OHIO RIVER TRAIL ON ROAD CONNECTOR, EDGEWORTH BOROUGH BEAVER STREET MODIFICATION PLAN

APRIL 2015

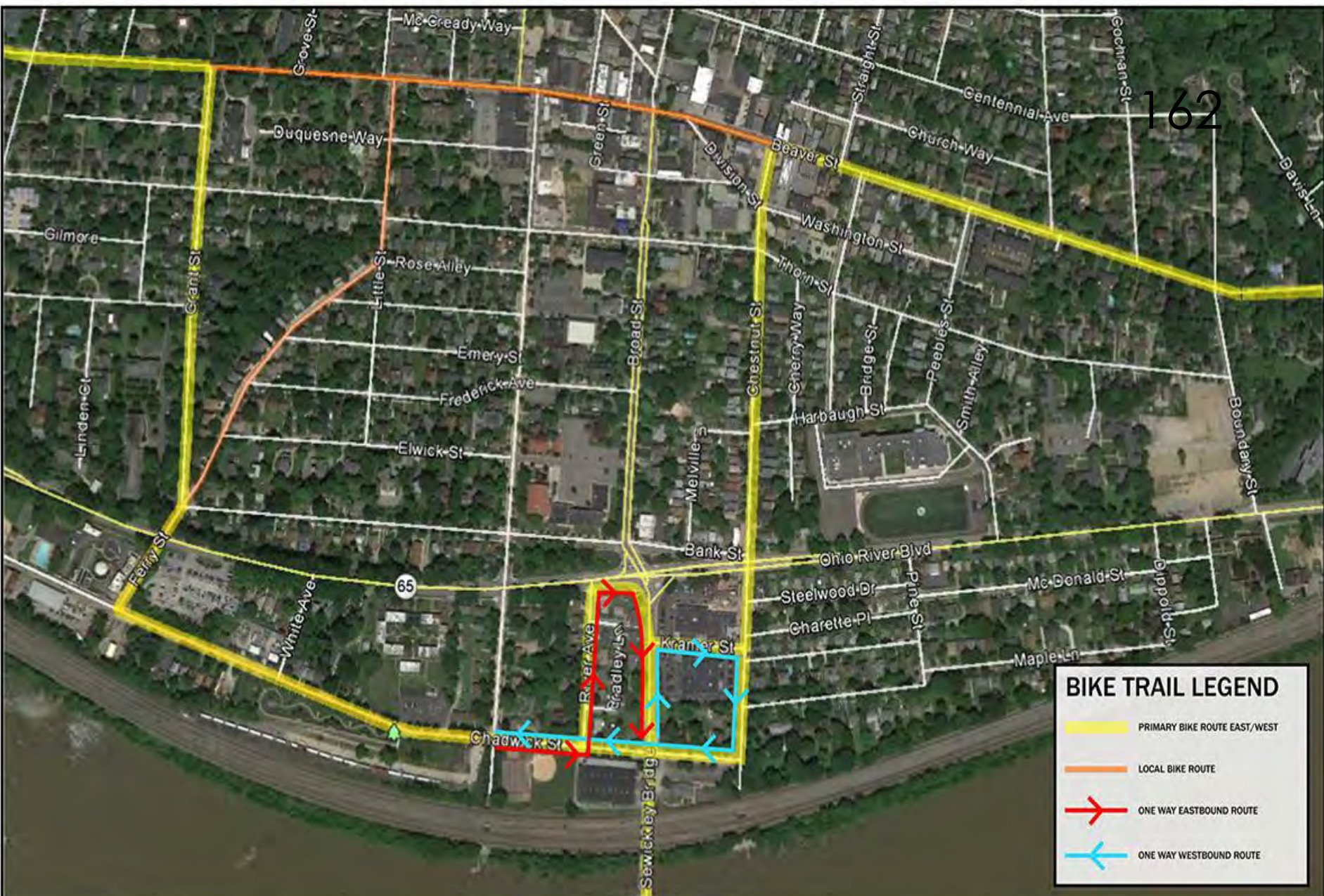


Ohio River Greenway Trail

Mackin Engineering

Phase 2

Sewickley



BIKE TRAIL LEGEND

- PRIMARY BIKE ROUTE EAST/WEST
- LOCAL BIKE ROUTE
- ONE WAY EASTBOUND ROUTE
- ONE WAY WESTBOUND ROUTE

EXISTING CONDITIONS

EXISTING CURB TO
CURB WIDTH = 34'

SHARE THE ROAD
SIGNAGE



SHARROWS; INSTALL IN
BOTH TRAVEL LANES



PROPOSED DRIVE AISLES = x2 @ 10'
PROPOSED SHOULDER = x2 @ 7'

OHIO RIVER TRAIL ON ROAD CONNECTOR, SEWICKLEY BOROUGH BEAVER STREET MODIFICATION PLAN



Ohio River Greenway Trail

Mackin Engineering

Phase 2

Glen Osborne

EXISTING CONDITIONS

EXISTING CURB TO
CURB WIDTH = 32'

ROAD EDGE
MARKING

SHARE THE ROAD
SIGNAGE

PROPOSED DRIVE AISLES = x2 @ 11'
PROPOSED SHOULDERS = x2 @ 5'

OHIO RIVER TRAIL ON ROAD CONNECTOR, GLEN OSBORNE BOROUGH BEAVER STREET MODIFICATION PLAN

EXISTING CONDITIONS

EXISTING CURB TO
CURB WIDTH = 27'-8"

SHARE THE ROAD SIGNAGE



ROAD EDGE MARKING

PROPOSED DRIVE AISLES = x2 @ 11'
PROPOSED SHOULDER = x1 @ 1.5'
PROPOSED SHOULDER = x1 @ 4'

OHIO RIVER TRAIL ON ROAD CONNECTOR, GLEN OSBORNE BOROUGH BEAVER STREET MODIFICATION PLAN

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Ohio River Greenway Trail Phase 1

Bob Genter, Director





Ohio River Greenway Trail

Mackin Engineering

Phase 1

Moon Township



Ohio River Greenway Trail

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Phase 1

Coraopolis

Mackin Engineering



Neville Island Bridge

- ▶ Three-way Connection
 - ❖ Ohio River Greenway Trail
 - ❖ Montour Trail
 - ❖ Three Rivers Heritage Trail

Mackin Engineering



Montour Trail

- ▶ Engineering and construction of the 0.8-mile extension of the Montour Trail is projected to begin in 2016.

CORAOPOLIS EXTENSION

PLANNED

Neville Is.
Exit 65

Coraopolis
Exit 64

CORAOPOLIS

MOON

Sewickley

↑
P
MONTOUR
ROAD

↑
P
HASSAM
ROAD

Hassam

Forest Grove

0

51 65

51

Bus.
60



Montour Street

4th Ave

**INTERSECTION IMPROVEMENTS AT NEVILLE
ISLAND BRIDGE AND 4TH AVE: CROSSWALKS &
ADA IMPROVEMENTS**

**CONSTRUCT NEW RAMP & EMBANKMENT
ALONG AT EXISTING RETAINING WALL;
10'-12' WIDTH TRAIL, 2-WAY TRAVEL**

**Montour Trail
Connector**

**Three Rivers
Heritage Trail
Connector**

**MONTOUR JUNCTION
SPORTS COMPLEX PROPERTY**

**EXTEND TRAIL BENEATH NEVILLE ISLAND
BRIDGE TO MONTOUR ST.; 10' WIDTH
TRAIL, 2-WAY TRAVEL**



Coraopolis to Pittsburgh's Southside Connector

Neville Island Bridge

51

Mackin



Mackin Engineering

Three Rivers Heritage Trail Feasibility Study

Bob Genter, Director





Ohio River Greenway Trail

Mackin Engineering

Coraopolis to Pittsburgh

Mackin Engineering – Three Rivers Heritage Trail



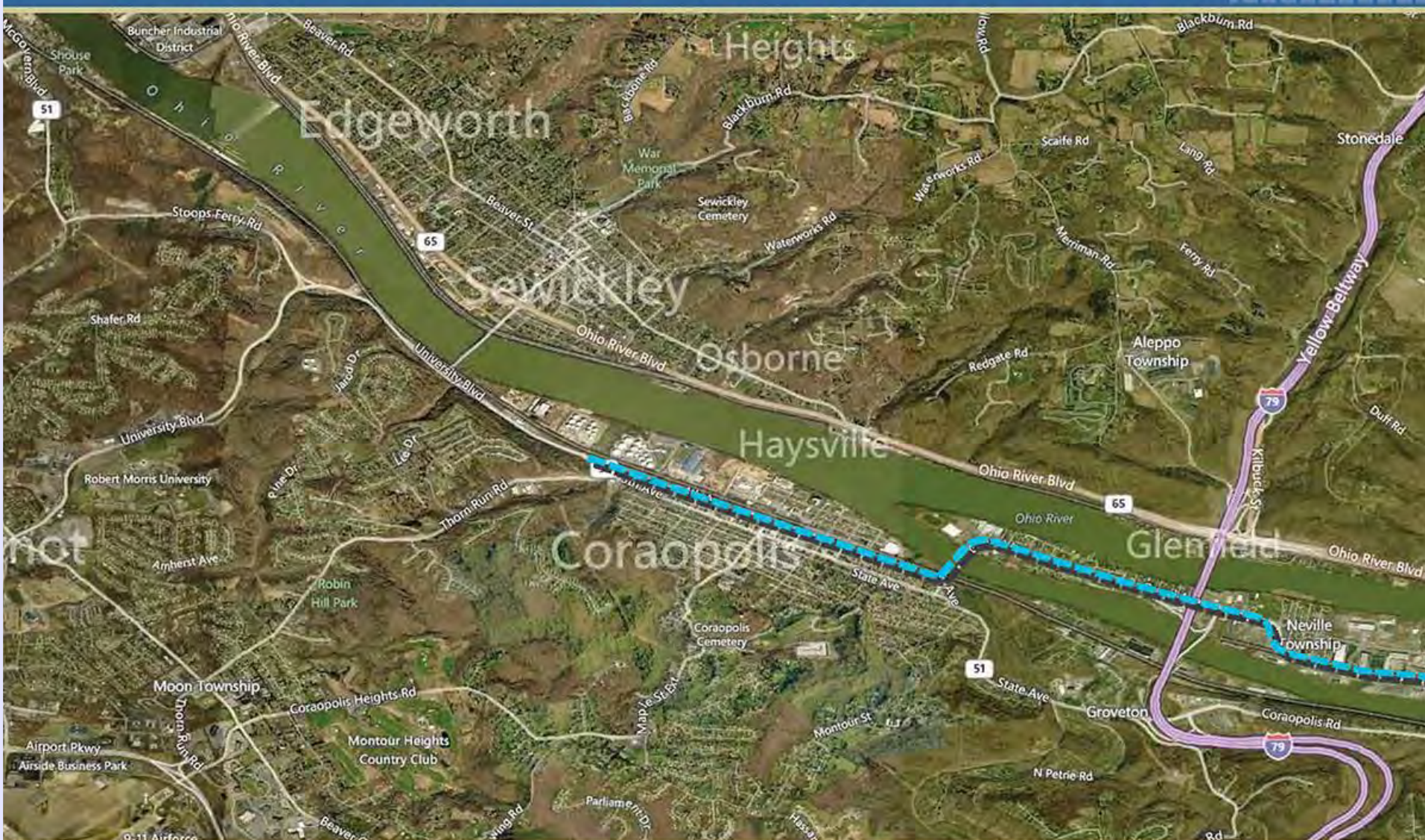
177

Coraopolis to Pittsburgh Feasibility Study

► Study Completed on June 10, 2013



Neville Island to Coraopolis Section



EXISTING CONDITIONS

BIKE ROUTE & "SHARE THE ROAD"
SIGNSSHARROWS; INSTALL CLEAR OF CAR DOORS.
DELINEATE EDGE OF ROAD PARKING SPACES AT 7' X 22'ENHANCEMENT RENDERINGS
5TH STREET BIKE ROUTE
CORAOPOLIS BOROUGH

FIGURE 15

180

ENHANCEMENT RENDERINGS

GRAND AVE. WEST OF I-79 INTERCHANGE

NEVILLE ISLAND

FIGURE 14



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RDC Park West
117 Industry Drive
Pittsburgh, PA 15275-1015
info@mackinengineering.com

BIKE ROUTE & "SHARE THE ROAD" SIGNS

EXISTING CONDITIONS

SHOULDER USED FOR ON-STREET PARKING;
BIKE LANE WOULD NOT BE FEASIBLE

SHARROWS; INSTALL IN RIGHT TRAVEL LANE



ENHANCEMENT RENDERINGS

NEVILLE ROAD
NEVILLE ISLAND

FIGURE 13

**BIKE LANE &
NO PARKING SIGNS**

EXISTING CONDITIONS

PEDESTRIAN ROUTE SYMBOL W/ DIRECTIONAL ARROW

**CONVERT EXISTING PAVED SHOULDERS TO
BIKE LANES, INSTALL PAVEMENT MARKINGS
(TYP.) BOTH SIDES OF ROADWAY**



INSTALL NO PARKING
SIGNS

BIKE ROUTE & "SHARE THE
ROAD" SIGNS

EXISTING CONDITIONS

'SHARROWS'; BOTH SIDES ROADWAY



ENHANCEMENT RENDERINGS
REJECTED ROUTE - ISLAND AVE.
EAST OF MCKEES ROCKS BRIDGE
STOWE TOWNSHIP

FIGURE 9



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RDC Park West
117 Industry Drive
Pittsburgh, PA 15275-1015
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BIKE ROUTE & "SHARE THE ROAD" SIGNS

EXISTING CONDITIONS

EXISTING ROAD WIDTH - 36'

'SHARROWS'; BOTH SIDES ROADWAY



184

ENHANCEMENT RENDERINGS

REJECTED ROUTE - ISLAND AVENUE AT BOUQUET STREET

STOWE TOWNSHIP

FIGURE 8

BIKE LANE SIGN

EXISTING CONDITIONS

EXISTING ROAD WIDTH - 36'

'ROAD DIET' & BIKE LANE CONDITION;
TWO CARTWAYS & 2-5' BIKE LANES/PAVEMENT
MARKINGS, EACH SIDE ROADWAY



185

ENHANCEMENT RENDERINGS
LOCAL LOOP ROUTE - SHOPPES AT CHARTIERS CROSSING
MCKEES ROCKS BOROUGH

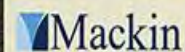
FIGURE 6



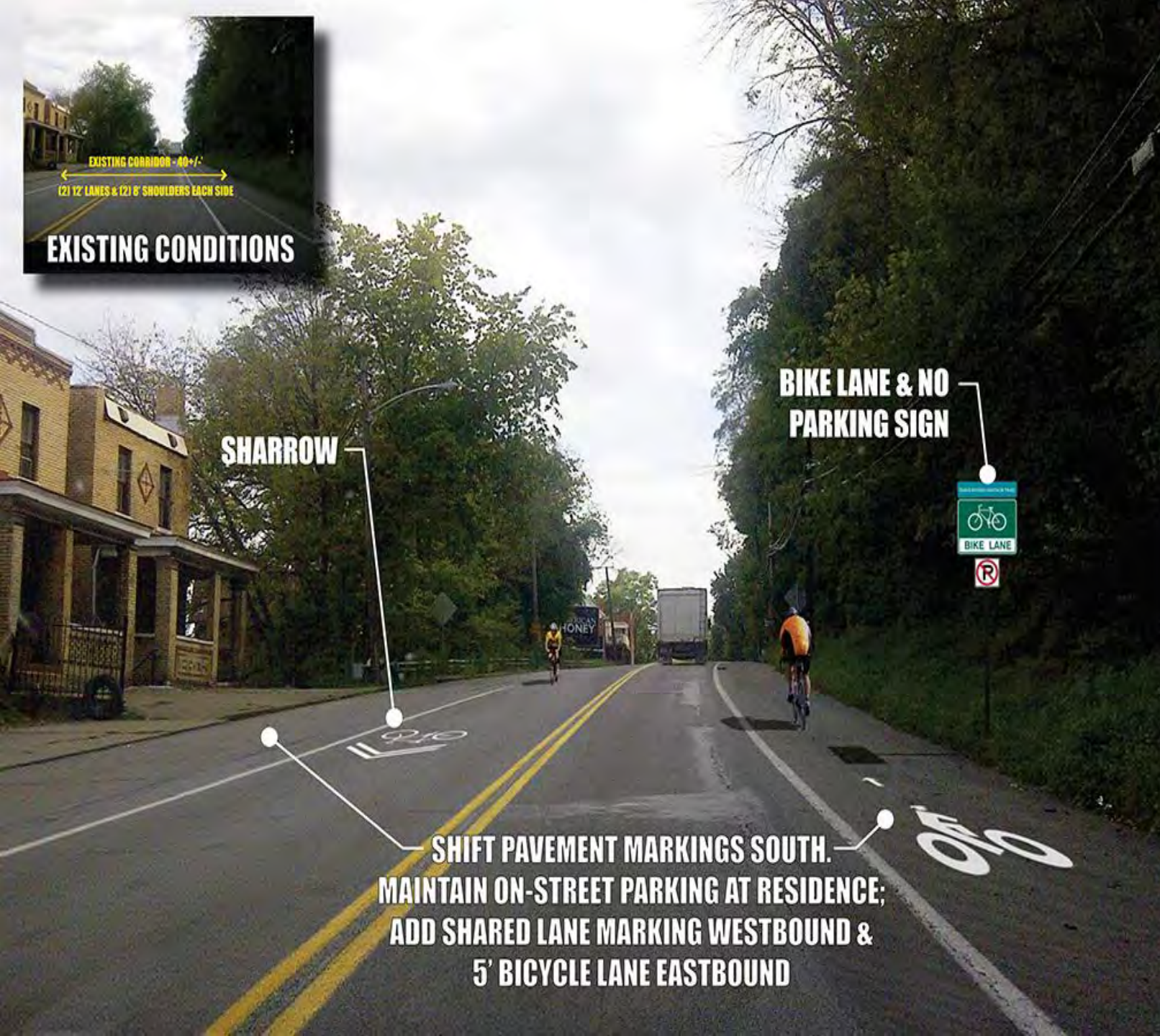
186

ENHANCEMENT RENDERINGS
ISLAND AVE. - EAST OF FLEMING BRIDGE
STOWE TOWNSHIP

FIGURE 12

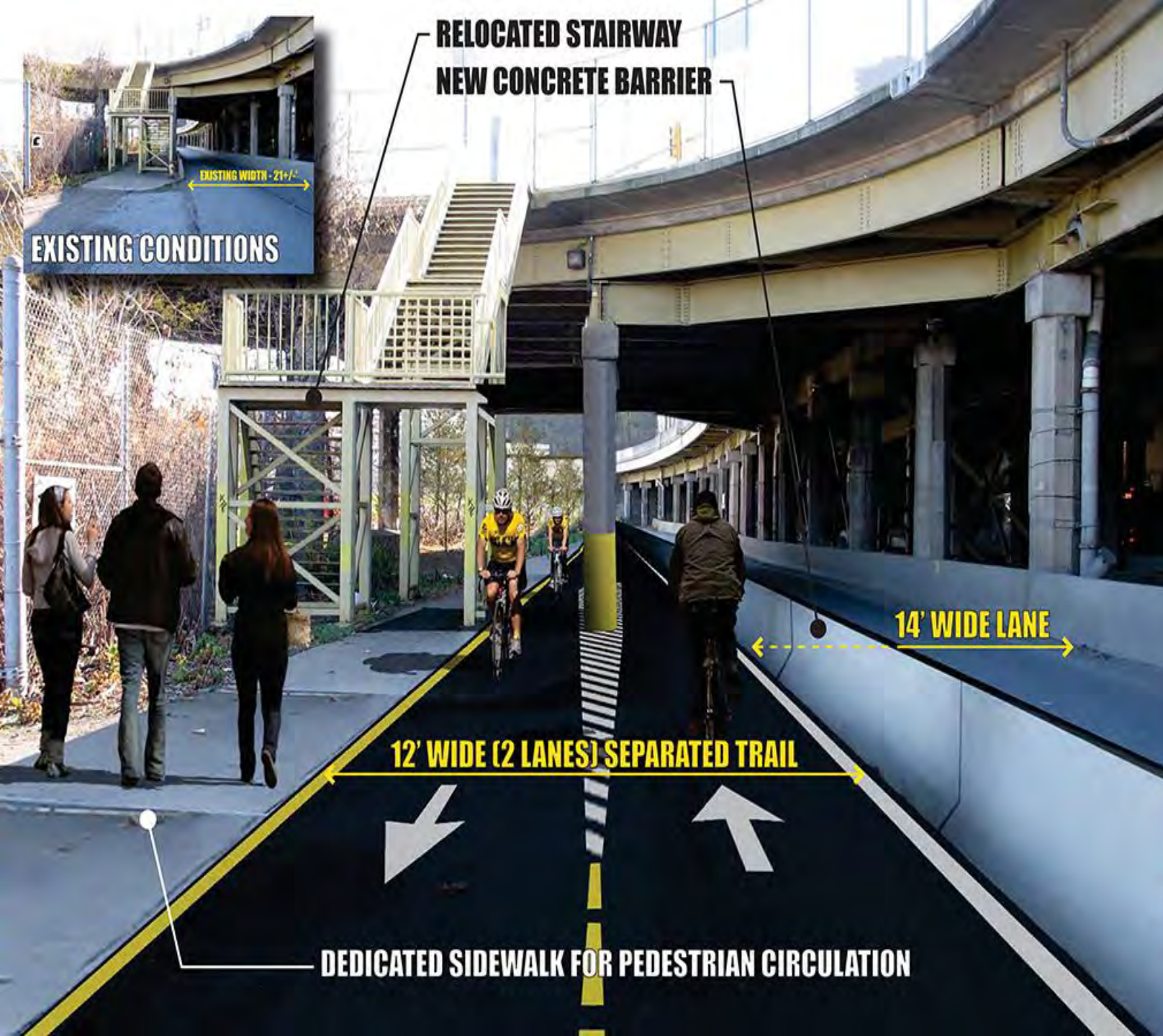


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117 Industry Drive
Pittsburgh, PA 15275-1015
info@mackinengineering.com



ENHANCEMENT RENDERINGS
RT. 51 - SEPARATED TRAIL
UNDER THE WEST END BRIDGE
CITY OF PITTSBURGH

FIGURE 1



Ohio River Trail Council Thank you!



*“Do not go where the path may lead,
go instead where there is no path and
leave a trail.”*

- Ralph Waldo Emerson

